

Software Tests Summary

We can almost never run a program on *every* possible input (there are far too many). So testing is really about **choosing a small, smart set of inputs** that still gives us confidence the program is correct. A **coverage criterion** is just a rule that tells you *which* inputs (or paths, or combinations) you must exercise — e.g. "run every line at least once" or "make every `if` go both true and false". Most of this guide is a tour of different such rules, how to satisfy them, and how they compare in strength. Keep asking two questions while you read: "*What must this criterion make me test?*" and "*How few tests can I get away with?*"

How to read this doc. Each section has: **Key definitions** (the vocabulary), **The recipe** (the mechanical steps to answer an exam question), a **Worked example**, **Exam patterns & gotchas** (traps that lose points), and a **Cheat sheet** (the compressed version to memorize). If a term looks unfamiliar, look for a > **Plain words**: note near its first use.

Topics

#	Topic	Exam frequency
1	Mutation Testing	Q1 every exam
2	Control Flow & Coverage Criteria	every exam
3	Data Flow Testing	most exams
4	Subsumption — Master Cheat Sheet	every exam
5	Combinatorial / Pairwise (AETG, IPO/IPOG)	every exam
6	Symbolic Execution	every exam
7	Concolic Testing (DART, CUTE)	most exams
8	FSM-based Testing (UIO, DS, W)	every exam
9	Black-box (ECP, BVA, Decision Tables, Domain)	sometimes
10	JUnit & Tooling (Pitest, JaCoCo)	support

1. Mutation Testing

Plain words

Mutation testing checks *how good your tests are* (not the program). The idea: deliberately break the program in tiny ways — each broken copy is a **mutant** — then see whether your test suite notices. A good suite should fail on a broken program. If a mutant slips through with all tests still passing, your tests have a blind spot. Think of it as "planting bugs on purpose to check that your bug-detector actually detects."

Key definitions:

- **Mutant** — a copy of program P with *one* small, still-compilable change, written P_i . "Syntactically-valid change" just means the edit still compiles/runs (you can't test a program that won't build). Examples: swap an operator ($> \rightarrow \geq$, $+$ $\rightarrow$ $-$), or change a boundary. One mutant = one tiny change.
- **Killed (dead) mutant** — at least one test in your suite T gives a *different output* on the mutant P_i than on the original P . That difference is the test "catching" the planted bug.
Survived — no test noticed the change (all tests give the same result on P_i as on P). Surviving mutants point at weak spots in your tests.
- **Equivalent mutant** — the change happens to make *no difference at all*: P_i and P behave identically on *every possible input* (e.g. rewriting $a*b$ as $b*a$). No test can ever kill it because there is nothing to catch — it's not really a different program. Deciding in general whether two programs always agree is **undecidable** ("undecidable" = no algorithm can answer it correctly for every case; you must argue each one by hand).
- **Mutation score** = $100 \times D / (N - E)$ — D = killed, N = total mutants, E = equivalent. It's the percentage of *killable* mutants your suite actually killed. Equivalents are removed from the **denominator** (they were never killable, so counting them would unfairly punish the suite), and never counted as killed. Higher score = stronger test suite.
- **Competent-programmer hypothesis** — the assumption that real programmers write *almost-correct* code, so real bugs are small slips (a wrong operator, an off-by-one), not wild rewrites. **Coupling effect** — the observation that tests catching these small planted bugs also tend to catch bigger, more complex bugs. Together these justify why testing with *single* tiny changes is worthwhile.
- A mutant survives if **either**: (1) no test even executes the mutated line, **or** (2) a test runs the line but the final *output* comes out the same anyway (this case includes equivalent mutants). "iff" = "if and only if".

The recipe (per mutant):

1. **Precondition**: run the suite T against the original P first — every test must pass. A failing test means **fix P and retest**, not a killed mutant. The whole process iterates: after adding tests, re-run until the score clears the chosen threshold.
2. **Locate** the mutated line and the exact change (e.g. $> \rightarrow \geq$).
3. **Find a reaching input** — an input that (a) actually runs the mutated line, AND (b) makes the mutated expression compute a different value there than the original would. Step (b) is called **infection**: the internal state is now "infected" (wrong) at that point. Just running the line isn't enough; the value has to actually diverge.

4. **Propagate:** an infected internal value is useless unless it changes what the program finally *outputs/returns* — that's **propagation** (the wrong value has to travel out to where a test can see it). Check the return value / output actually differs for that input. If for *every* input the output stays identical → the change never shows ⇒ **equivalent** (add to E). Otherwise it is **killable**.
5. **Killed?** A mutant is killed iff the *existing* test suite contains an input from step 2/3 whose asserted value now mismatches. If none → it **survives**.
6. **Count:** N = total mutants, E = equivalents, D = killed.
7. **Score** = $100 \cdot D / (N - E)$.
8. **To reach 100%:** for each surviving non-equivalent mutant, add a test whose input flips the mutated expression's outcome AND whose assertion checks the differing result.
Boundary-adjacent inputs (the two values straddling a comparison) kill the most mutants per test.

Worked example — one function, one test, four mutants. ($\text{foo}(x) = 2x$ if $x < 0$, else x .)

```
int foo(int x) {
2:  int y = 0;
3:  if (x < 0)          // C1
4:    y = -x;
5:  else
6:    y = x;
7:  if (x < 0)          // C2
8:    y = 2 * x;        // overwrites y when x<0
9:  return y;
}
```

The suite is a **single** test: `assertEquals(-2, foo(-1))` (passes on the original: $x=-1 \rightarrow$ line4 $y=1$, then line8 $y=-2$). Four mutants, applied one at a time:

#	mutator	change	on <code>foo(-1)</code>	verdict
M1	CONDITIONALS_BOUNDARY (line 3)	$x < 0 \rightarrow x \leq 0$	-2 (same)	equivalent
M2	MATH (line 8)	$2 * x \rightarrow 2 / x$	$2 / -1 = -2$ (same)	killable, survives
M3	INVERT_NEGS (line 4)	$y = -x \rightarrow y = x$	-2 (same)	equivalent
M4	NEGATE_CONDITIONALS (line 3)	$x < 0 \rightarrow x \geq 0$	-2 (same)	killable, survives

- **Which are killed by the test? None — all 4 survive** (each computes -2 on $x=-1$, matching the assertion). M2 survives only by luck: $2 * (-1)$ and $2 / (-1)$ both equal -2 .
- **Which are equivalent?**
 - **M1** — $x < 0$ vs $x \leq 0$ differ only at $x=0$; there original takes the else ($y=0$) and the mutant takes $y=-0=0$ — same output. No input distinguishes them.
 - **M3** — line 4 runs only when $x < 0$, but then line 8 ($x < 0$ true) **overwrites** $y=2*x$, so line 4's value is **dead**; changing $-x \rightarrow x$ never reaches the output.
- **Coverage of the one test?** `foo(-1)` only ever takes the $x < 0$ path, so the **else (line 6) never runs** $\Rightarrow$ statement coverage $< 100\%$, and each `if` takes only its True edge $\Rightarrow$ **branch coverage = $2/4 = 50\%$** . (Full coverage still wouldn't kill the equivalents — the weak-oracle point.)
- **Score:** killed $D=0$, total $N=4$, equivalent $E=2 \Rightarrow 100 \cdot D / (N-E) = 0 / (4-2) = 0\%$.
- **Add tests to kill the two killable ones $\rightarrow 100\%$:**
 - `assertEquals(-4, foo(-2))` **kills M2** ($2 * -2 = -4 \neq 2 / -2 = -1$).
 - `assertEquals(3, foo(3))` **kills M4** (original $3 \neq$ mutant -3) — and this hits the else branch, fixing coverage too.
 - Leave M1, M3 alone (equivalent — unkillable). New score $2 / (4-2) = 100\%$.

Exam patterns & gotchas:

- **Score formula:** memorize $100 \cdot D / (N-E)$. Equivalents leave the denominator; they are NOT killed.
- **Branch coverage $\neq$ mutants all killed:** 100% branch coverage does NOT guarantee killing all `CONDITIONALS_BOUNDARY` mutants — you cover both branches without testing the *boundary value* that distinguishes $>$ from $\geq$. Counterexample: `if (x > 0)` tested with $x=5$ and $x=-5$ covers both branches, but $x=0$ (where $>$ vs $\geq$ differ) is never tried $\rightarrow$ the mutant survives.
- **Writing a surviving mutant on purpose:** pick a comparison, test only inputs *far* from the boundary so the boundary swap doesn't change any asserted result.

Cheat sheet — Pitest DEFAULT mutators (*this is the group the exam draws from — memorize these; the exam really only uses the first three*):

Mutator (NAME)	What it does	Example
Conditionals Boundary (<code>CONDITIONALS_BOUNDARY</code>)	$< \leftrightarrow \leq, > \leftrightarrow \geq$	$a < b \rightarrow a \leq b$
Negate Conditionals (<code>NEGATE_CONDITIONALS</code>)	flip the whole relational op	$= \rightarrow ! =, > \rightarrow < =, \geq \rightarrow <$

Math (MATH)	swap arithmetic op	$\ast \rightarrow /$, $+\rightarrow -$, $\% \rightarrow \ast$, $<<\rightarrow >>$
Increments (INCREMENTS)	$++ \leftrightarrow --$	
Invert Negatives (INVERT_NEGS)	$-x \rightarrow x$	
Return Values	mutate returns: $\text{true} \leftrightarrow \text{false}$, $0 \rightarrow 1$, non-null $\rightarrow$ null	see version note below
Void Method Calls (VOID_METHOD_CALLS)	delete a void call	

Exact boundary table (memorize): $<\rightarrow <=$, $<=> <$, $>\rightarrow >=$, $>=> >$. Exact negate table: $==> !=$, $!=> ==$,
 $<=> >$, $>=> <$, $<\rightarrow >=$, $>\rightarrow <=$.

2. Control Flow & Coverage Criteria

Plain words

"Control flow" = the order in which statements run and the branch points (`if`, loops) that decide the route. We draw the program as a map — the **Control Flow Graph (CFG)** — and then pick coverage rules that say how thoroughly the map must be walked: every box? every fork-direction? every combination of conditions in a fork? The rules get progressively stronger (and need more tests). The recurring exam skills are: *draw the CFG*, *pick the smallest test set that satisfies a given rule*, and *say how many tests each rule needs*.

Key definitions:

- **CFG (Control Flow Graph)** — a diagram of the program's possible routes. Three node shapes: **computation** (rectangle = straight-line code that just runs top to bottom), **decision** (diamond = a condition, with a True edge and a False edge coming out), **merge** (circle = where two branches rejoin). Assume one **entry** and one **exit**; number every node so you can name paths like 1, 2, 3.
- **Statement (node) coverage** — every node runs at least once. This is the *weakest* rule (bare minimum: "no line of code went completely untested").
- **Branch (edge/decision) coverage** — every decision is taken **both** ways: each `if` goes True at least once and False at least once. **Subsumes statement** (see §4 — it forces every node to run too, so it's strictly stronger).
- **Basic-condition coverage** — in a compound condition like `a && b`, each *elementary* (atomic) part `a` and `b` is individually made both True and False at some point. *Incomparable with branch* (neither one guarantees the other — see §4 [CE2](#)).
- **Branch-and-condition** — satisfy branch coverage AND basic-condition coverage at the

same time.

- **Compound-condition** — test *every combination* of the atomic conditions in one decision. With N atoms that's up to 2^N combinations (short-circuit evaluation — where `&&/||` stops early once the result is decided — removes some impossible combinations).
- **MC/DC (Modified Condition/Decision Coverage)** — for *each* atomic condition, show it matters *on its own*: find two tests that differ in **only that one condition** and produce **opposite** overall decision results (proving that condition alone can flip the outcome). This needs about **$N+1$** tests for N conditions — far fewer than the 2^N of compound-condition — because each test is reused across several conditions. Required by aviation-safety standards **DO-178B / ED-12B**.
- **Cyclomatic complexity (McCabe, $v(G)$)** — a number measuring a module's **branching complexity** = the count of **linearly independent paths** through its CFG (equivalently, the size of a *basis set* of paths). Both course books define it as $v = e - n + 2$ (e = edges, n = nodes of the CFG; with P disconnected components it's $e - n + 2P$). Two equivalent shortcuts:
 - $v = (\text{\# binary decision points}) + 1$ — count each `if/while/for/case` and each `&&/||`; an `else` is **not** a decision. (Fastest by hand.)
 - v = number of enclosed regions of the planar CFG, +1. It measures *control structure, not size*, and equals the number of independent paths **cyclomatic testing** aims to cover. Rule of thumb (P&Y): <20 simple, >50 may be untestable.
 - *Worked example* (`calculateCyclomaticComplexity`):

```
while (y < 100) {           // decision 1
    if (y % 5 == 0) ...      // decision 2
    else if (y % 3 == 0) ... // decision 3
    else ...                // (else is NOT a decision)
}
```

3 binary decisions (`while`, `if %5`, `else-if %3`) $\Rightarrow v = 3 + 1 = 4$ (the same 4 you'd get from $e - n + 2$ after drawing the CFG).

- **Path (all-paths) coverage** — execute **every complete route from entry to exit** at least once. It is the **strongest** structural criterion (it subsumes all the others — cover every path and you cover every edge, node, and condition combination along the way), but it is usually **infeasible**: a single loop creates unboundedly many paths (0, 1, 2, ... iterations $\Rightarrow$ infinitely many tests), and even loop-free code with k independent decisions has up to 2^k paths. That blow-up is exactly why the weaker criteria (and, for loops, **boundary-interior** below) exist. When you *are* asked for a path-coverage test set (loop-free code only), list one input per feasible entry→exit path and mark any **infeasible** path (no input can drive it) rather than inventing one.
- **Infeasible (inexecutable) path** — a route that exists on the CFG diagram but **no input**

can ever actually make the program take, because the branch choices along it **contradict each other**. Simple example: a path that needs $x > 0$ at one `if` and later $x < 0$ at another `if`, with x never changed in between — no single x is both, so the path can never run. **How to detect it**: walk the path and AND together the condition each decision forces (True $\rightarrow$ the condition, False $\rightarrow$ its negation) into one big formula — the **path condition** — then ask "can any input satisfy all of these at once?" (this is exactly symbolic execution, §6). If the formula is **contradictory (unsatisfiable / UNSAT)** $\rightarrow$ the path is **infeasible**; if some assignment satisfies it $\rightarrow$ the path is **feasible** and that assignment is an input that drives it. Infeasible paths are why you can rarely reach 100% path- or du-path coverage: you **drop them and write "infeasible" (with the contradiction)** instead of inventing an input.

- **Boundary-interior** — a way to tame loops (which otherwise create infinitely many paths). It splits the loop paths into two classes. **Full coverage of each is about every subpath through the loop body, not just iteration count** — this is the part people get wrong:
 - **Boundary tests (full coverage)** = for **every distinct subpath through the loop body**, one feasible path that *enters the loop and exits after that single iteration* — **plus** the path that *skips the loop entirely*. Mechanically: unfold the CFG into a tree up to the **first repeated node** (the loop condition on its 2nd arrival), then stop and exit; **provide one feasible path per branch of that tree**. $\triangle$ One single-iteration path is **not** enough if the body has an `if` — you need *one per body-subpath*. In this course, this boundary set is the expected answer.
 - **Interior tests (full coverage)** = paths that iterate ≥ 2 times where the **first two iterations take different body-subpaths**; you enumerate the body's branch outcomes over iterations 1 and 2. Needs unfolding a **second** iteration, so the first-repeated-node tree does **not** produce them. The *more general* requirement — usually not required on the exam.
 - **Concrete `while(c){ if(d) X else Y }`**: the body has two subpaths (d -True $\rightarrow$ X, d -False $\rightarrow$ Y).
 - **Boundary (3 tests)**: skip loop (c false at once) · one iteration through **X** ($d=T$) · one iteration through **Y** ($d=F$).
 - **Interior (4 tests)**: two iterations with $d = \mathbf{TT, TF, FT, FF}$ (the TF/FT cases — where the iterations differ — are the ones that truly define "interior").
 - Quick test (course clarification): a feasible path starting with the unfolded prefix — *one iteration then exit* $\rightarrow$ *boundary*; *two-or-more differing iterations* $\rightarrow$ *interior*. Also aim for full **branch** coverage on any branches **outside** the loop.
- **Loop-boundary adequacy** — a simpler loop rule: run each loop **0 times, exactly 1 time, and more than 1 time** (the three qualitatively different loop behaviors).

⚠ **Loop-boundary vs boundary-interior — don't confuse them**

The word "boundary" means different things. Here "boundary" is a false friend: a *loop-boundary* "boundary" is an **iteration-count edge case** (0/1/many); a *boundary-interior* "boundary" test is a **path that barely enters the loop**.

- **Subsumption:** "A subsumes B" means A is at least as strong — any test set that satisfies A automatically satisfies B, for *every* program. (Full treatment in §4.)

The recipe:

Drawing the CFG: one node per basic block; each `if/loop` condition = a diamond with T/F edges; loop back-edge returns to the condition node; merge after branches. Number nodes; label which source line each represents (line numbers are required).

*Drawing a CFG that spans **two functions*** (caller + a callee you're told to include, e.g. `f()` calls `printProduct()`): **inline the callee's own CFG at the call site** — don't draw it as a single opaque node. At the call, add an edge **into the callee's entry**, draw the callee's full subgraph (its own diamonds/nodes), then an edge from the callee's **exit back to the caller's next node** — and if the callee **returns a value**, draw that return edge back to where the caller uses it (a classic lost point: forgetting the return-value edge). Number nodes continuously across **both** functions so you can name paths/du-paths over the whole thing.

Branch coverage + denominator: denominator = **number of outgoing edges from decision nodes** = $2 \times$ (number of decisions counted as branches). Example: two `ifs` → **4 branches**. Pick a minimal input set hitting each diamond's T and F.

Boundary-interior (the high-value recipe):

1. Draw CFG, identify the loop condition node.
2. **Unfold into a tree**, expanding until you reach a node you've *already visited* (the loop-condition node on its second arrival), then stop and exit the loop.
3. Enumerate every root-to-leaf subpath. **Always include the path that does NOT enter the loop** (condition false on first arrival).
4. For each subpath, find a *feasible* concrete input. If a prefix is infeasible (e.g. inner index can't exceed outer on iteration 1), state "infeasible" and continue.
5. Each path starts at entry, ends at exit. The set should be **minimal**.

Compound vs basic condition counting:

- Compound-condition tests for a single decision with N elementary conditions = up to 2^N . To "need >100": use **$N=7$** ($2^7=128 > 100$) elementary conditions in one decision.
- Basic-condition tests for that same decision = **2** (all-true row + all-false row suffice).

Building a minimal MC/DC set (list-the-table-then-pick; P&Y §12.5): MC/DC = *each* atomic condition must be shown to **independently flip the decision**. Target **$N+1$** tests for N conditions.

1. **List the full truth table** — all 2^N rows of the N atomic conditions.
2. **Mark infeasible rows** — cross out combos no input can produce (e.g. `income>50000=F`)

while `income>70000=T`; two conditions on the same variable). Do this **first** — you may not use an infeasible row in a pair.

3. **Compute the Decision** for every feasible row.
 4. **For each condition C_i , find an independence pair**: two feasible rows that differ in **only C_i** (all other conditions equal) and give **opposite** Decisions. The two rows *prove* C_i matters on its own.
 5. **Reuse rows to minimize**: one row can anchor several conditions' pairs (P&Y: "the same test case can cover several basic conditions"). Greedily pick anchors that serve multiple $C_i \rightarrow$ the set shrinks toward **$N+1$** .
 6. **Present the pairs**: list the chosen tests and, per condition, which two rows form its pair. The minimal set is **not unique** — any valid selection is accepted; short-circuit/coupling may force slightly more than $N+1$.
- *Don't-cares* (–) are allowed for a condition not evaluated (short-circuit) — it can shrink the table.

Worked example:

```
void checkNumbers(int[] numbers) {
  2: for (int i=0; i<numbers.length; i++)
  3:   if (numbers[i] % 2 == 0)           // even?
  4:     if (numbers[i] > 10)            // >10?
  5:       System.out.println(...);
  8: (exit)
}
```

CFG — nodes: **1** `i=0` · **2** `i<len?` (loop cond) · **3** `even?` · **4** `>10?` · **5** `print` · **6** `i++` · **8** `exit`. (T) / (F) = branch; node 6 is the loop back-edge to 2:

```

  1: i = 0
    |
    v
+--> 2: i < len?  --F--> 8: exit
    |
    | T
    | v
    | 3: even?  --F--> 6      (odd -> skip to i++)
    |   | T
    |   | v
    |   | 4: >10?  --F--> 6      (not >10 -> skip to i++)
    |   |   | T
    |   |   | v
    |   |   | 5: print  -----> 6
    |   |   |
+----- 6: i++  <-----+

```

Unfold to the first repeated node (2 on its 2nd arrival) → **minimal boundary-interior set (4 paths = skip + one per loop-body subpath):**

1, 2 (F) , 8	input [] -> loop not entered, exit
1, 2 (T) , 3 (T) , 4 (T) , 5, 6, 2 (F) , 8	input [12] -> even & >10 -> prints
1, 2 (T) , 3 (T) , 4 (F) , 6, 2 (F) , 8	input [2] -> even, not >10 -> nothing
1, 2 (T) , 3 (F) , 6, 2 (F) , 8	input [3] -> odd -> nothing

Exam patterns & gotchas:

- **Don't forget the loop-not-entered path** — always part of the boundary set.
- **Subsumption facts to quote** (all disproofs live in §4's [Counterexample library](#)):
 - boundary-interior **subsumes branch** (covering every subpath including in-loop branches covers every T/F edge).
 - branch **subsumes statement**; statement does NOT subsume branch.
 - branch does **NOT** subsume compound-condition → [CE1](#) (`if (a&&b)`, suite (T,T),(T,F): full branch, never tests (F,T)/(F,F)).
 - **basic-condition and branch are incomparable** → [CE2](#).
 - **loop-boundary and statement have NO subsumption either way** → [CE3](#) (both-directions counterexample).

Cheat sheet — criteria, denominators, subsumption:

Criterion	Obligation (denominator)	# tests (rough)	Subsumes
Statement	each node/basic block	—	(weakest)
Branch	each decision edge = 2 × #decisions	≥ #edges	statement
Basic condition	each elem. cond. T & F = 2 × #elem-conds	2 (often)	— (incomparable w/ branch)
Branch-and-condition	branch ∪ basic-condition		branch, basic-cond
Compound condition	every combination per decision = ≤ 2^N	up to 2 ^N	branch-and-condition
MC/DC	2 obligations per elem. condition	~N+1	branch-and-condition
Boundary-interior	each subpath of CFG unfolded to first repeated node	varies	branch

Loop-boundary	loop runs 0, 1, >1	3 per loop	— (incomparable w/ statement)
Path / all-paths	every path	∞ w/ loops	everything

Subsumption ladder (strong → weak): all-paths $\supset$ boundary-interior $\supset$ {MC/DC, compound-condition, cyclomatic, LCSAJ} $\supset$ branch-and-condition $\supset$ branch $\supset$ statement; basic-condition and loop-boundary sit at the base, *incomparable* to branch/statement respectively. (*This is the idealized, possibly-infeasible model — the one the exams use; see the note at §4's diagrams.*)

3. Data Flow Testing

Plain words

Control-flow testing cared about *which lines run*. Data-flow testing cares about *the life of each variable's value*: where it's **set** (given a value) and where it's later **used**. The worry is a broken link between them — e.g. code sets x but a bug means a stale or wrong x gets used downstream. So we pair up every "here's where x is set" with every "here's where that x is read", and require tests that actually travel from the set to the use. Vocabulary below is just names for "set" (**definition**), "read" (**use**), and "a route from set to read that doesn't overwrite x on the way" (**def-clear path**).

Key definitions:

- **Definition** $d_n(x)$ ("def"): x is *given a value* at node n — the left-hand side of $x = \dots$, a parameter receiving its argument at entry, or reading input into x . Parameters count as defined at the entry node.
- **Use** $u_n(x)$: x 's value is *read* (on the right-hand side of an assignment, inside a condition, or as a call argument). Two flavors:
 - **c-use (computation use)** — the value feeds a *computation* or output (an assignment, a `return`, a `print`); attached to a **NODE**. E.g. `return x+10` is a c-use of x .
 - **p-use (predicate use)** — the value is read *in a decision* (an `if`/loop condition). A p-use is attached to an **EDGE**, not a node — and a predicate node has **two** out-edges (True and False), so **one variable read in a condition creates two separate p-uses: one on the True edge, one on the False edge**. Each individual p-use is a single $(\text{variable}, \text{edge})$ pair. E.g. `if(flag)` gives p-use of `flag` on edge $\rightarrow \text{True}$ and p-use of `flag` on edge $\rightarrow \text{False}$ — two obligations. So "cover a p-use" = traverse **one** edge; "cover the condition's p-uses" (or **all-p-uses**) = traverse **both** edges.
- **def-clear path w.r.t. x** ("w.r.t." = with respect to): a route where none of the *in-between* nodes reassign or clear x . Meaning: the value set at the start is *still the same value*

when it reaches the end — the link is intact.

- **$d_m(x)$ reaches $u_n(x)$** : there exists a def-clear path from the def at m to the use at n — i.e. the value set at m can actually arrive, unchanged, at the use at n .
- **complete path** — a path that runs all the way from the **entry node to an exit node** (a whole execution route). This is what a **test case actually runs**: a du-path or def-clear path is usually only a *segment* of the program, so to exercise it you pick a complete path $entry \rightarrow \dots \rightarrow exit$ that *contains* that segment. (In the dataflow criteria, you "select complete paths which include the required def-clear paths.") **simple path** = no repeated nodes *except possibly the two endpoints* (so a single loop back to the start is allowed); **loop-free path** = all nodes distinct (no repeat at all).
- **du-path** (definition-use path, $n_1 \dots n_k$): a def-clear path from a *definition* of x to a *use* of x . Precisely: n_1 has a def of x , and **either** n_k has a c-use and the path is **simple** (no repeated nodes except possibly the endpoints), **or** the last edge has a p-use and the path up to it is **loop-free**.
- A node like $x = x+1$ is *both* a **use** of x (the old value, on the right) **and** a **def** of x (the new value, on the left) — order matters: it reads, then overwrites.

The recipe (mechanical):

1. **Draw the CFG**, number nodes, force a single entry/single exit (add an exit edge if a `return` dangles).
2. **Annotate each node**: list $d_i(var)$ for every assigned variable, and the uses.
Predicate node $\rightarrow$ p-use on **both** out-edges; assignment/return/print node $\rightarrow$ c-use in the node.
3. **Build the def-use table**: for every (def, use) pair of the same variable, find a def-clear path connecting them. Each such pair is one **obligation** — a thing some test must exercise. The full list of pairs is your obligation set (the checklist to tick off).
4. **Satisfy a criterion**: the criteria below differ only in *how many* of those def $\rightarrow$ use pairs you must cover. They range from lazy (`all-defs`: reach *some* use of each def) to thorough (`all-du-paths`: cover *every* route to *every* use). Read the table as "for each definition of x , how much must I cover?"

Criterion	Obligation per definition $d(x)$
all-defs	one def-clear path from each def to some (any one) use it reaches
all-c-uses	a def-clear path from each def to every c-use it reaches
all-p-uses	a def-clear path from each def to every p-use it reaches — i.e. to BOTH out-edges (True and False) of every decision the def reaches ($\Rightarrow$ all-p-uses subsumes all-branches)

all-c-uses/ some-p-uses	all c-uses; if a def reaches no c-use, then at least one p-use
all-p-uses/ some-c-uses	all p-uses; if a def reaches no p-use, then at least one c-use
all-uses	one (any) def-clear path to every use (all c-uses AND all p-uses) — one route per use suffices
all-du-paths	every def-clear du-path (cycle-free / simple-cycle) to every use — <i>all</i> routes, not just one $\Rightarrow$ trivially includes all-uses; may be exponential

1. **Feasibility check:** drop infeasible paths; you rarely hit 100%.

Worked examples — the two dataflow subsumption disproofs (both live in §4's [Counterexample library](#), so all subsumption counterexamples sit in one place):

- **CE4 — full branch coverage $\not\supseteq$ all-defs:** a two-`if` program where both tests give 100% branch coverage yet the def of `x` never reaches one of its uses. This is the canonical "all-defs is easy to break" trap (bullet below).
- **CE5 — all-c-uses/some-p-uses $\Leftrightarrow$ all-p-uses/some-c-uses:** two small programs (one per direction) proving the pair **incomparable**. Direction 1 is the `foo(x, y)` program: suite `{1-2-4-6}` covers the only c-uses but skips p-use edges. Key fact used: **all-p-uses requires both out-edges of every decision**, so a single path can never satisfy it.

Exam patterns & gotchas:

- p-use lives on the EDGE (count both T and F edges); c-use lives in the NODE. A predicate node has **no defs** by assumption.
- A node like `x = x + 1` is simultaneously a **c-use** and a **def** of `x`; that def kills earlier defs through it.
- "all-defs" only needs **some** use per def — easy to satisfy, easy to break with branch coverage.
- all-c-uses and all-p-uses are **incomparable**.
- For "/some" criteria: the "some" clause only fires when a def reaches **zero** uses of the other kind.
- all-du-paths can be exponential; when asked to "list all du-paths," include both branches around a diamond.
- **Variable appears in two functions:** if you're asked for the du-paths of `x` and `x` is defined/used in **both** functions (e.g. a caller and the callee you inlined into the CFG — see §2), **list the du-paths in each function**. Don't report only one; a `def→use` pair in the callee and one in the caller are separate obligations (they don't cross the function boundary unless a value is passed and the CFG is genuinely inlined).

Cheat sheet — criteria table (slide CFG $1:d(x) \rightarrow \{2:u(x), 3:u(x)\} \rightarrow 4 \rightarrow \{5:u(x), 6\}$):

Criterion	Requires	Satisfying path(s)
all-defs	$d_1(x)$ to <i>some</i> use	1, 2, 4, 6
all-uses	$d_1(x)$ to u_2, u_3, u_5	1, 2, 4, 5, 6 + 1, 3, 4, 6
all-du-paths	every cycle-free du-path	1, 2, 4, 5, 6 + 1, 3, 4, 5, 6

4. Subsumption — Master Cheat Sheet

Plain words

"Subsumption" ranks coverage criteria by strength. Saying **A subsumes B** means: *if you've satisfied A, you've automatically satisfied B* — A is the tougher bar, so passing it gets B for free. (Example: branch coverage subsumes statement coverage — take every `if` both ways and you can't help but run every line.) The exam skill is almost always the *opposite* direction: **disprove** a claimed subsumption by inventing one small program + one test suite that satisfies A yet misses B. This section is the toolkit for that.

Key definitions:

- **A subsumes B** ("A includes B"): for **every** program P, **every** test suite that satisfies A on P also satisfies B on P. A is then *strictly stronger*. (Note the "for every program" — a single program where A implies B is *not* enough; it must always hold.)
- **Equivalent**: A subsumes B *and* B subsumes A (they demand the same coverage).
Incomparable: neither subsumes the other (each can be satisfied while missing something the other requires).
- Caution: subsumption is a *logical* relation only — "A is stronger on paper". It does **not** guarantee A finds more real bugs in practice.

⚠ **An example is NOT a proof.** Because "A subsumes B" is a $\forall$ -program claim, the two directions need opposite evidence.

The recipe — to disprove "A subsumes B", find ONE program P + ONE suite T with: T satisfies A on P, but T does NOT satisfy B on P.

1. Pick the obligation B requires that A does **not**.
2. **Build a tiny program** where that exact obligation can be isolated — a single extra statement, branch, def-use pair, or loop iteration A can skip.
3. **Construct the smallest suite T** meeting all of A's obligations while deliberately avoiding B's distinguishing obligation.
4. **Verify both claims explicitly**: (a) T satisfies A (enumerate A's obligations, show each

met); (b) T misses ≥ 1 of B's obligations (name it).

5. For "no subsumption in BOTH directions" (incomparability), repeat with a **second** program/suite swapping roles.

Counterexample library

*This is the one place every non-subsumption / incomparability in the doc is proved, each with **one program + one suite**. Other sections point here by ID (CE1–CE6). Template for all of them: show T satisfies A (every A-obligation met), then name the single B-obligation T misses.*

CE1 — branch $\not\supseteq$ compound-condition (equivalently, branch $\not\supseteq$ "all 2^N combinations").

```
if (a && b) X; else Y;
```

Suite { (a,b)=(T,T), (T,F) }: the decision is **True** then **False** $\Rightarrow$ **full branch coverage**. But compound-condition needs all $2^2=4$ atom-combinations, and (F,T), (F,F) are never tried $\Rightarrow$ **branch $\not\supseteq$ compound-condition**. (Referenced from §2.)

CE2 — basic-condition $\Leftrightarrow$ branch (incomparable, both directions).

Direction 1 — branch $\not\supseteq$ basic-condition. Program `if (a || b) X; else Y;` with suite { (T,F), (F,F) }: decision **True** then **False** $\Rightarrow$ full branch, yet atom **b is never True** $\Rightarrow$ basic-condition unmet. *Direction 2 — basic-condition $\not\supseteq$ branch.* Program `if (a && b) X; else Y;` with suite { (T,F), (F,T) }: each atom takes **both** T and F $\Rightarrow$ basic-condition met, yet the decision is **False in both tests** (the true-branch is never taken) $\Rightarrow$ branch unmet. Neither direction holds $\Rightarrow$ **incomparable**. (Referenced from §2.)

CE3 — statement $\Leftrightarrow$ loop-boundary (incomparable, both directions).

```
1 int foo(int x, int y) {
2   while (x > 0)
3     x--;
4   if (y == 0)
5     return x;
6   return y; }
```

Direction 1 (loop-boundary adequate, NOT statement adequate): suite `foo(0,0)` (loop 0 $\times$), `foo(1,0)` (1 $\times$), `foo(2,0)` (>1 $\times$). All keep `y==0` $\Rightarrow$ **statement 6 never executed** $\Rightarrow$ loop-boundary $\not\supseteq$ statement. *Direction 2 (statement adequate, NOT loop-boundary adequate):* suite `foo(0,0)` (loop 0 $\times$, hits line 5) + `foo(1,1)` (loop 1 $\times$, hits line 6). **Every statement** executed, but loop **never runs >1** $\Rightarrow$ statement $\not\supseteq$ loop-boundary. The two are **incomparable**. (Referenced from §2.)

CE4 — full branch coverage $\not\supseteq$ all-defs (dataflow).

```
1 int f(int a, int b) {
2   int x = 0;          // def D1 of x
```



```

3  if (a > 0)
4      x = 1;          // def D2 of x  (overwrites D1 on this path)
5  if (b > 0)
6      print(x);       // the only use of x
7  }

```

Suite { **(a=1,b=1), (a=-1,b=-1)** } takes both edges of each `if` $\Rightarrow$ **full branch coverage**. But **D1 (x=0)** can reach the use only via $a \leq 0$ (so it isn't overwritten) **and** $b > 0$ (so line 6 runs) — and neither test does both: the first overwrites `x` (D2), the second keeps `x=0` but skips the use. So **D1 reaches no use $\Rightarrow$ all-defs fails** at 100% branch coverage $\Rightarrow$ branch $\not\supseteq$ all-defs. (Referenced from §3.)

CE5 — all-c-uses/some-p-uses $\Leftrightarrow$ all-p-uses/some-c-uses (incomparable, dataflow).

Reminder (textbook): a p-use of a variable in a predicate is associated with **each out-edge**, so **all-p-uses forces both the T and F edge of every decision** (that's why all-p-uses $\supseteq$ all-branches). A single path can therefore never satisfy all-p-uses. Each direction below needs its **own** program.

Direction 1 — all-c-uses/some-p-uses $\not\supseteq$ all-p-uses/some-c-uses.

```

void foo(int x, int y) {          // 1: def x, def y
    if (x > 0 && y < 0) {          // 2: p-use x,y — edges 2->3 (T), 2->4 (F)
        if (x > 10) return;       // 3: p-use x — edges 3->5 (T), 3->6 (F)
    } else {
        print(x, y);              // 4: c-use x,y
    }
}                                  // 5: return, 6: exit

```

Suite { **1-2-4-6** } (one test with $x \leq 0$, so the outer `if` is false $\rightarrow$ `print`). The only c-uses are `x, y` at node 4, reached def-clear $\Rightarrow$ **all-c-uses/some-p-uses satisfied** (`x` and `y` have c-uses, so the criterion asks no p-use of them). But all-p-uses needs *both* edges of every predicate — $2 \rightarrow 3, 3 \rightarrow 5, 3 \rightarrow 6$ are never taken $\Rightarrow$ **all-p-uses/some-c-uses fails**.

*Direction 2 — all-p-uses/some-c-uses $\not\supseteq$ all-c-uses/some-p-uses (needs a def that gets **killed** on the paths p-use coverage happens to use):*

```

void bar(int v, int w) {          // 1: def v, def w
    if (v > 0) { }                 // 2: p-use v — edges 2->3 (T), 2->4 (F)
    else { v = 1; }               // 4: v=1  $\rightarrow$  KILLS v (node 3 = empty T-branch)
    if (w > 0) { v = 2; }          // 5: p-use w — edges 5->6 (T), 5->7 (F); 6: v=2  $\rightarrow$ 
    else { }                      // 7 = empty F-branch
    print(v);                     // 8: c-use v
}

```

Suite { **(v>0,w>0): 1-2-3-5-6-8, (v≤0,w≤0): 1-2-4-5-7-8** } takes both edges of both predicates $\Rightarrow$ **all-p-uses/some-c-uses satisfied** (`v` and `w` have p-uses). But the c-use of the *original* `v` (def@1)

needs a def-clear path to node 8, which exists **only** via node-2-True **and** node-5-False (1-2-3-5-7-8, i.e. $v > 0 \wedge w \leq 0$) — a corner neither test hits (each test kills v at node 4 or 6 first) $\Rightarrow$ that c-use pair is missed $\Rightarrow$ **all-c-uses/some-p-uses fails**.

Both directions $\Rightarrow$ **incomparable**.

CE6 — full branch coverage $\nsubseteq$ all-c-uses (dataflow).

```
1 int f(int w, int y) {
2   int x, r;
3   if (w < 0) x = 1; else x = 2;          // TWO defs of x (one per arm)
4   if (y < 0) r = x + 10; else r = 99; // c-use of x ONLY on the y<0 arm
5   return r; }
```

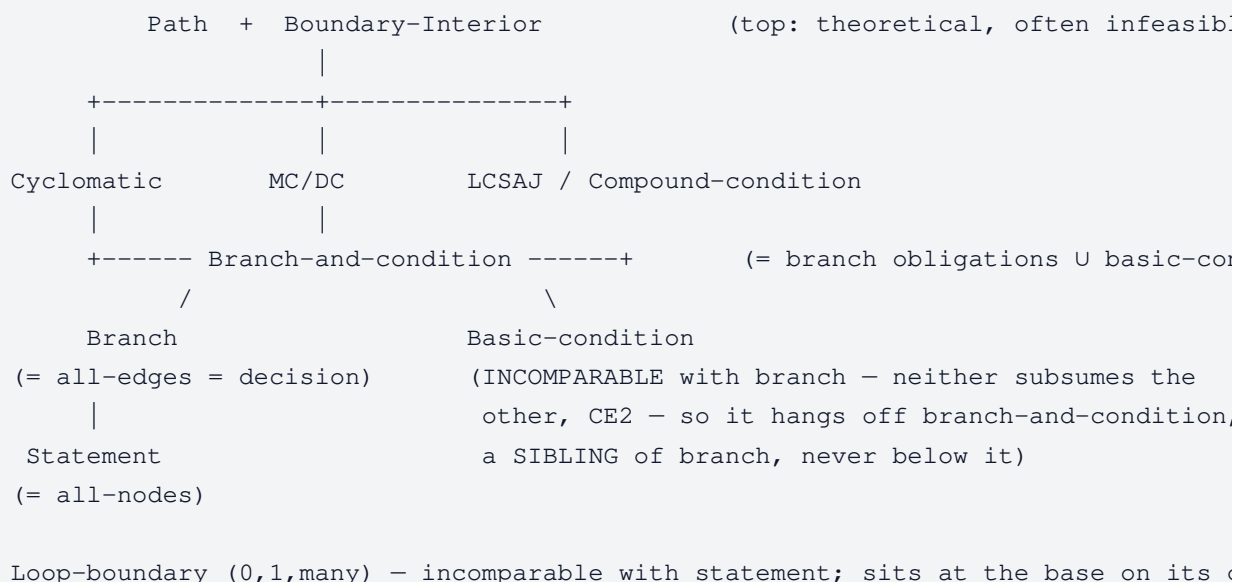
x is defined on both arms of the first `if`, and has one **c-use** ($x + 10$) on the true arm of the second `if`. all-c-uses therefore has **two** obligations: *def-from-the- $w < 0$ -arm $\rightarrow$ the c-use* and *def-from-the- $w \geq 0$ -arm $\rightarrow$ the c-use*. Tests **{w=-1, y=1}** and **{w=1, y=-1}** take both T and F of each `if` $\Rightarrow$ **full branch coverage**. But the first test defines x on the $w < 0$ arm and then takes $y \geq 0$ (never reaching the c-use), while the c-use is only reached by the second test, which defined x on the $w \geq 0$ arm $\Rightarrow$ the pair *def-on- $w < 0$ -arm $\rightarrow$ c-use* is **never exercised** $\Rightarrow$ all-c-uses unmet $\Rightarrow$ branch $\nsubseteq$ all-c-uses.

Exam patterns & gotchas:

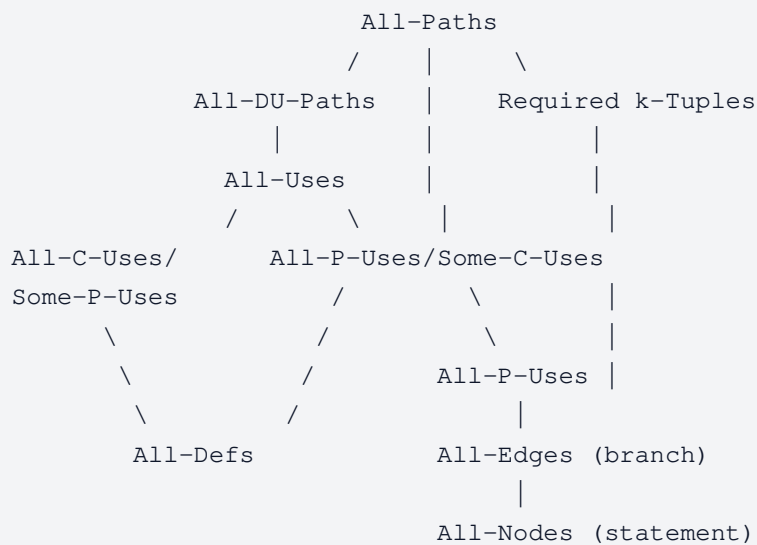
- Branch = "all-edges"; statement = "all-nodes"; decision $\equiv$ branch.
- **Boundary-interior subsumes branch.**
- **Loop-boundary (0,1,many) is at the BASE** — incomparable with statement ([CE3](#)); do not confuse with boundary-interior.

Cheat sheet (A $\rightarrow$ B means "A subsumes B", i.e. A stronger):

STRUCTURAL hierarchy:



DATAFLOW hierarchy (Rapps–Weyuker "includes"); top = strongest:



5. Combinatorial / Pairwise Testing (AETG & IPO/IPOG)

Plain words

Suppose a feature has several settings (parameters), each with a few possible values — say $\text{Table} \in \{\text{Coffee, Desk, Kitchen}\}$, $\text{Color} \in \{\text{Brown, White, Red}\}$, $\text{Size} \in \{\text{Small, Medium}\}$. Testing *every* combination is $3 \times 3 \times 2 = 18$ tests here, and explodes fast with more parameters. The insight behind **pairwise (2-way) testing**: most bugs are triggered by *one* setting or the *interaction of two* settings, rarely by three-plus at once. So we don't need every full combination — we only need every **pair** of values (from any two parameters) to appear together in *at least one* test. That collapses the suite dramatically (often to a handful of tests) while still catching the vast majority of interaction bugs. **AETG** and **IPO/IPOG** are two algorithms that build such a small test set.

What you're GIVEN and what you PRODUCE

- **Given:** a list of **parameters**, and for each one its **set of allowed values** (its "domain").
⚠ **Parameters can have any number of values, and different counts each** — e.g. P1 has 3 values, P2 has 3, P3 has 2. This is normal and the exams test it deliberately. There is **no special formula** for the multi-valued case — the mechanics below are identical; the only effect of a bigger domain is *more pairs to cover* and *uneven pair counts* when you count.
- **Produce:** a small set of **tests** (each test = one value chosen for *every* parameter) such that for **every pair of parameters**, **every** combination of one value from each appears in **at least one** test.

Key definitions.

- **t-way / pairwise (t=2):** for every group of t parameters, every value-combination of those t appears in ≥ 1 test. Pairwise is the $t=2$ case (every *pair*). "At least once" — it does **not** need to be balanced/equal counts.
- **Pair:** a specific (value-of-Pi, value-of-Pj) with $i \neq j$. E.g. (Table=Coffee, Size=Small). Pairs are always across **two different** parameters.
- **π (pi) = the set of pairs still *uncovered*.** This is your running checklist / bookkeeping object. The moment a test covers a pair, cross it off π . **You stop when π is empty** — every pair covered. Keeping π correct is where most exam marks are won or lost.
- **Covering array & orthogonal array — the mental model.** Picture your test set as a **table: one row per test, one column per parameter**, each cell holding a value. Both "arrays" are just names for such a table with a coverage guarantee about its columns:
 - **Covering array $CA(N; t, k, v)$** — a table of N rows (tests), k columns (parameters), each cell one of v values, such that: *pick any t columns, and every combination of their values shows up in **at least one** row*. For pairwise ($t=2$): **every pair appears ≥ 1 time** — that's the minimum we actually want. Reading the notation: N =#tests, t =strength (2 = pairwise), k =#parameters, v =#values per parameter. Its size grows only **logarithmically in the number of parameters** — why pairwise scales so well. (When parameters have *different* value counts, people write $CA(N; t, v_1 v_2 \dots v_k)$ or a "mixed-level" array; the idea is unchanged.)
 - **Orthogonal array $L_{\text{Runs}}(\text{Levels}^{\text{Factors}})$** — the *stronger, balanced* version: **pick any two columns, and every combination appears **exactly the same number of times**** (that fixed count is the array's "index", usually 1). Reading the notation $L_4(2^3)$: **4 runs** (rows/tests), **3 factors** (columns/parameters), each with **2 levels** (values) — the ³ is the number of columns, the 2 is the values-per-column. So $L_8(2^7) = 8$ tests, 7 binary parameters.
 - **How they relate:** *every orthogonal array is also a covering array, but not vice versa*. "Appears exactly-equally" (orthogonal) is a tighter demand than "appears at least once" (covering). The price of that balance: orthogonal arrays are **rigid** — they exist only for special sizes (e.g. value counts that are prime powers, equal-sized domains) and are often **bigger** than the smallest covering array for the same job. So we usually *build* covering arrays (via AETG/IPO); an orthogonal array is a nice ready-made table that's sometimes **handed to you as a ready-made starting set** (see "Orthogonal-array as a starting set" below).
 - **Concrete $L_4(2^3)$** (3 binary parameters, values 1/2):

run	P1	P2	P3
1	1	1	1
2	1	2	2
3	2	1	2

Check any two columns — e.g. P1 & P3: the pairs (1,1),(1,2),(2,1),(2,2) each appear **exactly once** $\Rightarrow$ orthogonal (balanced). It's automatically a covering array too (each pair appears $\geq$ once). A covering array *only* needs that " $\geq$ once" — so for larger problems it can skip rows an orthogonal array would be forced to keep for balance.

- **Covering array beats orthogonal — worked contrast (4 binary parameters).** Now take **4** binary parameters (values 0/1). Exhaustive testing = $2^4 = 16$ tests. An **orthogonal** array must have runs divisible by 4 for *every* column-pair to be balanced, and no orthogonal array exists for 4 binary factors in 4 runs — the smallest is **L8 = 8 runs** (it actually holds up to 7 factors). But a **covering** array (each pair $\geq$ once, balance not required) does the job in just **5 tests**:

run	P1	P2	P3	P4
1	1	1	1	1
2	1	0	0	0
3	0	1	0	0
4	0	0	1	0
5	0	0	0	1

Verify all $\binom{4}{2} = 6$ column-pairs — e.g. P1&P4: (1,1) in run 1, (1,0) in run 2, (0,0) in runs 3–4, (0,1) in run 5 $\Rightarrow$ all four combos present. The same holds for every other pair (each of the 6 pairs gets its 00/01/10/11 exactly as run 1 supplies the 11, the "one-hot" runs supply the 10/01, and the zero-heavy runs supply 00). So **5 < 8 < 16**: the covering array is smaller than the orthogonal array precisely because it drops the exactly-equally-often demand and keeps only "appears at least once." (Balance costs the extra 3 rows and buys nothing for pair *coverage*.)

AETG (Automatic **E**fficient **T**est **G**enerator) — **the recipe** (*builds one complete test at a time, greedy*):

1. **Build π** = every pair across every parameter-pair. Count = $\sum_{\{i < j\}} |P_i| \cdot |P_j|$.
(Optional binary convention: start with all-0s / all-1s tests first and remove their pairs.)
2. **Repeat until π is empty.** Each pass builds exactly ONE new test:
3. **Pick the first (parameter, value):** the one appearing in the **most remaining pairs of π** . In practice: count, for each parameter-value, how many uncovered pairs still contain it (an "occurrence count" table), and take the max. Ties $\rightarrow$ first in listed order.
4. **Generate m candidate tests.** Each candidate fills the *remaining* parameters in some **order** (given by the question, or random). m is a setting you choose (e.g. $m=1$ or $m=3$).
5. **Greedy per-parameter fill:** going through that candidate's order, for each next

parameter choose the **value that forms the most pairs still in π with the values already fixed so far**. $\triangle$ Only look **back** at already-assigned parameters, never ahead. Ties $\rightarrow$ first value. (*Multi-valued changes nothing here — you simply have more values to try; count each and take the max.*)

6. **Score each finished candidate** = total pairs in π it covers (re-count over the *whole* test).
7. **Keep the max-score candidate** (ties $\rightarrow$ first); add it as a test, remove all its pairs from π . Back to step 2.

Sub-skill: "list all pairs to add when extending to a new parameter" (a very common AETG/IPO sub-question). "You already have a pairwise set covering P_1, P_2 ; now add a new parameter P_3 — list all pairs that must be covered." **Answer = only the pairs that involve the new parameter** (the P_1 – P_2 pairs are already done, don't re-list them). That is: every value of $P_3 \times$ every value of each existing parameter.

Formula: pairs to add = $\sum$ over each existing P_j of $(|P_j| \times |P_3|)$. **Example (multi-valued):** $P_1=\{\text{Coffee, Desk}\}$ (2), $P_2=\{\text{Brown, White, Red}\}$ (3), new $P_3=\{\text{S, M, L}\}$ (3) $\rightarrow P_1 \times P_3 = 2 \times 3 = 6$ pairs + $P_2 \times P_3 = 3 \times 3 = 9$ pairs = **15 pairs** to add.

$\triangle$ **This counts *pairs*, not *tests***. All 15 are distinct ($P_1 \times P_3$ pairs use P_1 's values, $P_2 \times P_3$ pairs use P_2 's values — nothing overlaps), so you can't lower the 15. But the number of *tests* needed is far smaller, because **one test covers several pairs at once**: a row (*Coffee, Brown, S*) knocks out (*Coffee, S*) *and* (*Brown, S*) together. That reuse is exactly what IPO horizontal growth does — append a P_3 value to an existing row and it covers one $P_1 \times P_3$ pair and one $P_2 \times P_3$ pair simultaneously.

IPO (In-Parameter-Order; the t -way generalization is **IPOG, IPO-General**) — **the recipe** (*adds one parameter at a time; deterministic*):

The core idea: start with a table that's already pairwise-correct for the **first two** parameters, then bring in each new parameter one at a time. Adding a parameter is done in two moves — **grow the table sideways** (widen the rows you already have) first, and only if that leaves pairs uncovered, **grow it downward** (add new rows). Sideways is free coverage (no new tests); downward is the last resort.

1. **Initialization:** write out the **full** cross-product of the first two parameters — every (P_1 -value, P_2 -value) combination, one per row. (With only two parameters each row *is* a pair, so there's no way to do fewer; this is $|P_1| \times |P_2|$ rows.)
2. **For each next parameter P_i (P_3 , then P_4 , ...):**
 - **a. Build π** = all the new pairs this parameter introduces = every (value of an earlier parameter P_j , value of P_i) — exactly the "pairs involving the new parameter" from the sub-skill above. (Earlier parameters are already mutually covered; only pairs *touching* P_i are new.)
 - **b. Horizontal (sideways) growth — widen existing rows, add NO new rows.** Go down the existing rows top to bottom and **append one P_i -value** to

each. For a given row, appending value b covers, in one shot, the pair of b with *every* earlier parameter's value already in that row — so pick the b that covers the **most pairs still in π** (ties $\rightarrow$ first listed value). Cross those pairs off π before moving to the next row. You have exactly as many rows as before, so at most $\#_{\text{rows}}$ distinct P_i -values get placed here; if P_i has more values (or more pairs than rows can absorb), some pairs are left for step c.

- **c. Vertical (downward) growth — add new rows for the leftovers.** Any pair still in π after horizontal growth needs a fresh row. For each leftover pair $(P_j=a, P_i=b)$: **first try to reuse** an existing vertical-growth row — one whose P_j slot is already a **or** blank *and* whose P_i slot is already b **or** blank — and fill in its blanks. **Only if none fits, add a brand-new row** with $P_j=a, P_i=b$, and ***** (**don't-care = "any value"**) in every other column. Reusing rows before adding new ones is what keeps the suite small.
- **d.** Replace any leftover $*$ with any valid value, then move on to the next parameter P_{i+1} (its horizontal growth now runs over *all* rows, including the ones vertical growth just added).

Worked example A — multi-valued AETG (uneven domains), full run. $P_1=\{C,D\}$ (2), $P_2=\{B,W,R\}$ (3), $P_3=\{S,M\}$ (2). Conventions (state them in your answer): **$m=1$** (one candidate per test); fill the still-unassigned parameters in **index order** $P_1 \rightarrow P_2 \rightarrow P_3$; all ties (first pick and value) break to the **first-listed** value/parameter.

Build π — $|P_1| \cdot |P_2| + |P_1| \cdot |P_3| + |P_2| \cdot |P_3| = 6+4+6 = 16$ pairs:

P_1P_2 : (C,B) (C,W) (C,R) (D,B) (D,W) (D,R) P_1P_3 : (C,S) (C,M) (D,S) (D,M) P_2P_3 : (B,S) (B,M)

Test 1.

- **Count $\rightarrow$ first pick:** $C \rightarrow 5, D \rightarrow 5, S \rightarrow 5, M \rightarrow 5$, each of $B/W/R \rightarrow 4$. Tie at 5 $\rightarrow$ first $\rightarrow$ **$P_1 = C$** .
- **Fill P_2 ($P_1=C$):** (C,B),(C,W),(C,R) all =1 $\rightarrow$ tie $\rightarrow$ **B**.
- **Fill P_3 (C,B):** $S \rightarrow (C,S)_{p13} + (B,S)_{p23} = 2; M \rightarrow 2 \rightarrow$ tie $\rightarrow$ **S**.
- $\Rightarrow$ **Test 1 = (C,B,S)**, score 3; remove (C,B),(C,S),(B,S) $\rightarrow$ **13 left**.

Test 2.

- **Count $\rightarrow$ first pick:** on 13 left: $D \rightarrow 5, M \rightarrow 5, W \rightarrow 4, R \rightarrow 4, C \rightarrow 3, S \rightarrow 3, B \rightarrow 2$. Tie D vs M $\rightarrow$ first parameter $\rightarrow$ **$P_1 = D$** .
- **Fill P_2 ($P_1=D$):** (D,B),(D,W),(D,R) all =1 $\rightarrow$ tie $\rightarrow$ **B**.
- **Fill P_3 (D,B):** $S \rightarrow (D,S)_{p13} = 1$ [(B,S) gone]; $M \rightarrow (D,M)_{p13} + (B,M)_{p23} = 2 \rightarrow$ **M**.
- $\Rightarrow$ **Test 2 = (D,B,M)**, score 3; remove (D,B),(D,M),(B,M) $\rightarrow$ **10 left**.

Test 3.

- **Count $\rightarrow$ first pick:** $W \rightarrow 4$ [(C,W)(D,W)+(W,S)(W,M)], $R \rightarrow 4$, else ≤ 3 . Tie W vs R $\rightarrow$ **P_2**

= W.

- **Fill P1** ($P2=W$): $(C,W)=1, (D,W)=1 \rightarrow \text{tie} \rightarrow \mathbf{C}$.
- **Fill P3** $(C,W): S \rightarrow (W,S)_{p23}=1 [(C,S) \text{ gone}]; M \rightarrow (C,M)_{p13}+(W,M)_{p23}=2 \rightarrow \mathbf{M}$.
- $\Rightarrow$ **Test 3 = (C,W,M)**, score 3; remove $(C,W),(C,M),(W,M) \rightarrow \mathbf{7 \text{ left}}$: $P1P2:(C,R)(D,W)(D,R) \cdot P1P3:(D,S) \cdot P2P3:(W,S)(R,S)(R,M)$.

Test 4.

- **Count** $\rightarrow$ **first pick**: $R \rightarrow 4 [(C,R)(D,R)+(R,S)(R,M)]$ is the max $\rightarrow \mathbf{P2 = R}$.
- **Fill P1** ($P2=R$): $(C,R)=1, (D,R)=1 \rightarrow \text{tie} \rightarrow \mathbf{C}$.
- **Fill P3** $(C,R): S \rightarrow (R,S)_{p23}=1 [(C,S) \text{ gone}]; M \rightarrow (R,M)_{p23}=1 \rightarrow \text{tie} \rightarrow \mathbf{S}$.
- $\Rightarrow$ **Test 4 = (C,R,S)**, score 2 (only $(C,R),(R,S)$; (C,S) already covered) $\rightarrow \mathbf{5 \text{ left}}$: $P1P2:(D,W)(D,R) \cdot P1P3:(D,S) \cdot P2P3:(W,S)(R,M)$.

Test 5.

- **Count** $\rightarrow$ **first pick**: $D \rightarrow 3 [(D,W)(D,R)+(D,S)] \rightarrow \mathbf{P1 = D}$.
- **Fill P2** ($P1=D$): $(D,W)=1, (D,R)=1 \rightarrow \text{tie} \rightarrow \mathbf{W}$.
- **Fill P3** $(D,W): S \rightarrow (D,S)_{p13}+(W,S)_{p23}=2; M \rightarrow 0 \rightarrow \mathbf{S}$.
- $\Rightarrow$ **Test 5 = (D,W,S)**, score 3; remove $(D,W),(D,S),(W,S) \rightarrow \mathbf{2 \text{ left}}$: $P1P2:(D,R) \cdot P2P3:(R,M)$.

Test 6.

- **Count** $\rightarrow$ **first pick**: $R \rightarrow 2 [(D,R)+(R,M)] \rightarrow \mathbf{P2 = R}$.
- **Fill P1** ($P2=R$): $(D,R)=1 \rightarrow \mathbf{D}$.
- **Fill P3** $(D,R): M \rightarrow (R,M)_{p23}=1 \rightarrow \mathbf{M}$.
- $\Rightarrow$ **Test 6 = (D,R,M)**, score 2; π **empty**.

Answer — 6 tests (vs $2 \times 3 \times 2 = 12$ exhaustive): $(C,B,S), (D,B,M), (C,W,M), (C,R,S), (D,W,S), (D,R,M)$. Verify: $P1P2$ gets all 6, $P1P3$ all 4 $[(C,S)t1,(C,M)t3,(D,S)t5,(D,M)t2]$, $P2P3$ all 6. ✓ Takeaway: multi-valued is purely mechanical — bigger domains just mean more values to count and more pairs to clear; the not-every-test-scores-3 rows (t4, t6) are normal near the end.

When $m > 1$

The run above uses $m=1$ (one candidate per test). With $m > 1$ the only change is the "generate candidates" step: at each test you build m candidates (each with its own fill order), **score every candidate over its whole finished test**, and keep the highest (ties $\rightarrow$ first). Larger $m \rightarrow$ more candidates per step $\rightarrow$ usually **fewer total tests**, at more computation.

Worked example B — IPOG (full trace: init $\rightarrow$ horizontal $\rightarrow$ vertical). Three parameters: $P1=\{1,2\}, P2=\{1,2\}, P3=\{1,2,3\}$. ($P3$ has 3 values, so horizontal growth *can't* place them all in the 4 existing rows — that's what forces vertical growth, the part exams love to test.)

Step 1 — Initialize with the $P1 \times P2$ cross-product (4 rows):

row	P1	P2
1	1	1
2	1	2
3	2	1
4	2	2

Step 2 — Add P3. π = the 12 pairs touching P3: $P1 \times P3 = (1,1)(1,2)(1,3)(2,1)(2,2)(2,3)$ and $P2 \times P3 = (1,1)(1,2)(1,3)(2,1)(2,2)(2,3)$ (read each as (Pj-value, P3-value)).

2b — Horizontal growth (append a P3 value to each existing row; pick the value covering the most still-uncovered pairs, counting against **both** P1 and P2 in that row):

row	P1 P2	try P3=1	P3=2	P3=3	pick	pairs removed from π
1	1 1	$(P1:1, P3:1) + (P2:1, P3:1) = 2$	2	2	1 (tie $\rightarrow$ first)	$(1,1)_{p1},$ $(1,1)_{p2}$
2	1 2	1	$(1,2)_{p1} + (2,2)_{p2} = 2$	2	2	$(1,2)_{p1},$ $(2,2)_{p2}$
3	2 1	1	$(2,2)_{p1} + (1,2)_{p2} = 2$	2	2	$(2,2)_{p1},$ $(1,2)_{p2}$
4	2 2	$(2,1)_{p1} + (2,1)_{p2} = 2$	0	2	1 (tie $\rightarrow$ first)	$(2,1)_{p1},$ $(2,1)_{p2}$

Rows after horizontal growth, and what's **left in π** (4 pairs — all the P3=3 pairs, since value 3 never got placed):

1	1	1	π left = { (P1:1, P3:3), (P1:2, P3:3), (P2:1, P3:3), (P2:2, P3:3) }
1	2	2	
2	1	2	
2	2	1	

2c — Vertical growth (each leftover pair needs a row; reuse a * row before adding a new one):

- $(P1=1, P3=3) \rightarrow$ no rows yet, add **row 5 = (1, *, 3)**.
- $(P2=1, P3=3) \rightarrow$ row 5 has P2=* and P3=3 already $\Rightarrow$ **fill the blank**: row 5 = (1, 1, 3).
- $(P1=2, P3=3) \rightarrow$ row 5 has P1=1 (fixed, $\neq 2$), can't reuse $\Rightarrow$ add **row 6 = (2, *, 3)**.
- $(P2=2, P3=3) \rightarrow$ row 6 has P2=* and P3=3 $\Rightarrow$ fill: row 6 = (2, 2, 3).

π is now empty. **Final suite — 6 tests** (vs $2 \times 2 \times 3 = 12$ exhaustive):

row	P1	P2	P3
1	1	1	1
2	1	2	2
3	2	1	2
4	2	2	1
5	1	1	3
6	2	2	3

Sanity-check one pair-type: $P2 \times P3 \rightarrow (1,1) r1, (2,2) r2, (1,2) r3, (2,1) r4, (1,3) r5, (2,3) r6 \Rightarrow$ all six present. ✓ The two takeaways: horizontal growth did the bulk of the work "for free" (no new rows for values 1 & 2), and vertical growth added the minimum rows for value 3, **reusing row 5 before opening row 6**.

Special-parameter twists (recurring "adapt the algorithm" sub-questions):

- **Fault-prone parameter** — "each value of P3 must appear $\geq$ twice with every other value": change π construction — **put every pair that involves P3 into π twice**; leave the other pairs at multiplicity one. Run growth/greedy normally, but **remove only ONE copy** of a doubled pair each time a test covers it — so the pair must be covered twice before it leaves π . (Works for both IPO and AETG).
- **Critical parameter** — "(P2,1) must appear in $\geq 75\%$ of tests": this is a *frequency* constraint, not a pair constraint, so **don't fiddle with π counts** (you don't know the final test count in advance). Instead, in AETG: when choosing the first (param,value) of each new test, if (P2,1) is currently in $< 75\%$ of tests so far, **force-select it**; and after all pairs are covered, keep **adding redundant tests containing (P2,1)** until the 75% threshold is met.
- **Orthogonal-array as a starting set**: if you're handed an orthogonal array (or any set of prebuilt tests), use it as the **starting tests**: build the full pair list, **strike out every pair those starting tests already cover**, then run AETG/IPO only on what's left $\rightarrow$ far fewer iterations.
 - **Seed IPO with an $L_n(2^{(k-2)})$ OA** : an OA is already a minimal pairwise-covering set for the parameters it contains, so $L_n(2^{(k-2)})$ gives you n rows that **already cover all pairs among $k-2$ of the k boolean parameters — for free**. So instead of IPO's usual Step 1 (start from the full $P1 \times P2$ cross-product), **use the OA's n rows as the initial test set**, then run IPO's parameter-addition (**horizontal $\rightarrow$ vertical growth**) for the remaining $k - (k-2) = 2$ parameters — building π from **only the pairs that involve those 2 new parameters** (the OA already covers the rest). You skip covering the first $k-2$ parameters entirely.

△ **Why you can't just *duplicate* an OA's columns to fake more parameters** (a classic "why doesn't this work?"). Tempting shortcut: you have $L_4(2^3)$ covering 3 parameters and you want 6, so you copy the 3 columns to the right ($P4:=P1, P5:=P2, P6:=P3$):

run	P1	P2	P3	P4	P5	P6	(P4=P1, P5=P2, P6=P3)
1	1	1	1	1	1	1	
2	1	2	2	1	2	2	
3	2	1	2	2	1	2	
4	2	2	1	2	2	1	

The trap is the pair **between a column and its own clone** — e.g. P1 & P4. Because P4 is *identical* to P1 in every row, that column-pair only ever shows the **matching** values (1, 1) and (2, 2); the **mismatched** pairs (1, 2) and (2, 1) can **never** appear. So pairwise coverage is broken for P1–P4, P2–P5, and P3–P6. (Cross pairs like P1–P5 are fine — they're just the original P1–P2 pairs again.) Duplication buys columns but not coverage; you must genuinely extend the array (add the missing pairs via AETG/IPO), not clone it.

Exam patterns & gotchas.

- **m=1 vs m=3**: larger $m \rightarrow$ more candidates per step $\rightarrow$ **fewer total tests** but **more computation**. $m=1$ is fast but yields a bigger suite. AETG is **non-deterministic** (random orders); IPOG is **deterministic**.

Cheat sheet — AETG vs IPOG:

	AETG	IPO / IPOG
Unit added per step	one complete test	one parameter
Strategy	greedy, m candidates, keep best	init $\rightarrow$ horizontal $\rightarrow$ vertical growth
Determinism	non-deterministic	deterministic
Complexity	higher	lower
Flexibility	—	extend existing set; * don't-cares
Setting	$m = \text{\#candidates}$ (bigger $\rightarrow$ fewer tests, slower)	—

6. Symbolic Execution

Plain words

Instead of running the program on *actual numbers*, run it on *symbols* that stand for "any input". As you walk one path through the code, you track two things: **PV** = what each variable now equals *in terms of those symbols* (e.g. `c1 = x*x`), and **PC** = the list of conditions the inputs must satisfy to have taken this exact path (e.g. `x > y`). At the end, the PC is a set of equations; if a solver can find numbers satisfying it, those numbers are a real test input that drives this path — and if the PC is contradictory (unsatisfiable), the path is **impossible** and needs no test. This is how you prove things like "this ERROR line can never be reached".

Key definitions.

- **Symbolic value** — instead of a concrete number, each input is given an uppercase symbol standing for "whatever the caller passes" (`x→X`, `arr→A`, its length `→ SIZE_OF_A`). Literal constants (like `1`, `0`) stay as themselves.
- **Symbolic state / PV (Program Variables)** — the current value of every variable written as a formula in those symbols. An **assignment** updates PV (e.g. after `c1 = x*x`, PV has `c1 = x*x`); it never touches the PC.
- **Path condition (PC)** — the running list of branch conditions joined by AND (`/\`), recording what must be true to follow this path. Taking a branch's **True** side appends the condition; taking the **False** side appends its **negation** (`!condition`).
- **Feasibility / SAT** — a path is *feasible* (a real input can follow it) exactly when its PC is **satisfiable**. You hand the PC to a constraint solver, which answers **SAT** (+ a concrete example — "here are numbers that work"), **UNSAT** (no numbers can satisfy it), **UNKNOWN**, or **TIMEOUT**. `UNSAT` $\Rightarrow$ path infeasible $\Rightarrow$ that code is unreachable via this path.
- **Reaching ERROR** — to check whether a specific `ERROR` line can run, AND together the conditions of exactly the branches on the route to it, and test that PC for satisfiability.

The recipe — columns `line` | **PV** | **PC**:

1. **Entry**: bind every parameter to its symbol in PV. PC empty.
2. **Assignment**: update only PV (substitute current symbols), e.g. `c1 = x*x`.
3. **Branch**: nothing in PV; append constraint to PC with `/\`. **Convention: take the FALSE branch first**, then negate that last constraint to flip to True on the next run. Simplify and say so (`X != X+1  $\equiv$  TRUE`, `X == X+1  $\equiv$  FALSE`).
4. **Return/ERROR**: write symbolic return value in PV, or mark `ERROR`.
5. After each run: state the **symbolic return value** and the **full PC**; then negate the last branch and re-run.

Worked example — unreachable ERROR:

Specify the PC that reaches the error in this code. Is it reachable?

```
void foo(double x){ c1 = x*x; if (c1+1==0){ if (c1-1==0){ ERROR; }}
```

line	PV	PC
entry	$x = X$	
c1	$c1 = X * X$	
if1		$X * X + 1 == 0$
if2		$X * X + 1 == 0 \wedge X * X - 1 == 0$

- **Full PC to ERROR:** $X * X + 1 == 0 \wedge X * X - 1 == 0$.
- **Reachable? NO.** $X * X + 1 == 0$ forces $X * X = -1$, impossible for real X (would need imaginary i); also both $v+1==0$ and $v-1==0$ can't hold at once. Canonical **unreachable-error** pattern.
- **Branch-coverage denominator:** two ifs $\Rightarrow 2 \times 2 = 4$.

Worked example — false-branch-first:

```
runSymbolic(int x,int y){ if(x>y) x=y+1; else y=x+1; if(x==y) ERROR; return x+y; }
```

Run 1 (false branch first $\Rightarrow x \leq y$, line $y=x+1$ runs):

line	PV	PC
entry	$x=X, y=Y$	
if1(F)		$X \leq Y$
$y=x+1$	$y = X+1$	
if2(F)		$X \leq Y \wedge X \neq X+1 \equiv X \leq Y$
return	$2 * X + 1$	

Symbolic return $2 * X + 1$, PC $X \leq Y$. Negate last $\rightarrow$ aim at ERROR: PC becomes $X \leq Y \wedge X \neq X+1 \equiv \text{FALSE} \Rightarrow$ **infeasible**, ERROR unreachable on this branch.

Exam patterns & gotchas.

- **Branch denominator** = $2 \times \# \text{decisions}$ (loop conditions count, infeasible also counted).
- **Unreachable errors:** spot UNSAT PCs — $x * x + 1 == 0, v + 1 == 0 \wedge v - 1 == 0, x > 0 \wedge y > 0 \wedge x + y < 0$. Answer "not reachable" + algebraic reason; never invent an input.
- **Satisfying input:** if the PC is SAT, give one concrete tuple that satisfies it (for PC $x \leq y$, answer $x=0, y=0$). **Finding an array out-of-bounds bug is the same skill:** an access

`arr[b+1]` is only safe while $0 \leq b+1 \leq \text{SIZE_OF_A} - 1$, so to *hit* the bug you add the violating constraint $b+1 > \text{SIZE_OF_A} - 1$ to the PC and solve. E.g. `arr` has 4 slots (indices 0–3, `SIZE_OF_A` = 4) and the code reads `arr[b+1]`: solving $b+1 > 3$ gives $b = 3$, which reads index 4 — one past the end $\Rightarrow$ out-of-bounds.

- **What changes in symbolic execution to guarantee MC/DC coverage?** Normally a compound decision is treated as **one** branch — you add the whole condition (or its negation) to the PC and fork just True/False. The change:
 1. For each decision, **first build its MC/DC cases** — the $\sim N+1$ rows: all-true, then flip **one** atom at a time (each flip must flip the overall result).
 2. **Run one symbolic execution per MC/DC row**, and to the PC add **each atomic condition's required truth value separately** (not the compound condition as a single unit). *Example* $a \wedge b \wedge c \rightarrow 4$ MC/DC rows $\Rightarrow$ **4 runs**, each PC forcing that row's atom values: $a/\backslash b/\backslash c$, $!a/\backslash b/\backslash c$, $a/\backslash !b/\backslash c$, $a/\backslash b/\backslash !c$.
 3. **Fallback:** if MC/DC is impossible for a condition (e.g. an atom that can't independently flip the outcome — masked or coupled), handle it "**as usual**" = the ordinary behavior from the intro line: **fork the whole decision True/False** (add $c / \neg c$ to the PC), with **no** per-atom MC/DC constraints. (You just get decision/branch coverage for it.)
 - **Per-atom or per-decision?** "if for a particular *condition* full MC/DC is impossible" is ambiguous — **state your reading**:
 - *Per basic condition (theoretically correct, what real tools do):* keep the independence pairs for the atoms that **can** be shown independent; drop **only** the impossible atom's obligation (ordinary handling for it) $\rightarrow$ "partial MC/DC".
 - *Per decision (the exam's likely simplifying intent, since "full MC/DC coverage" is a decision-level notion):* if the decision can't be **fully** MC/DC-covered, revert the **whole** decision to ordinary True/False — don't attempt partial MC/DC.
- **To get the next path, negate the *last* constraint of the previous run's PC** (not an earlier one), then re-solve. Walking $x \leq y \wedge x \neq x+1 \rightarrow$ flip the last $\rightarrow x \leq y \wedge x == x+1$. Always the most recently added conjunct.
- **Path-exploration ORDER (DFS + backtracking, "false-first").** "Give the first k paths a symbolic engine explores" If code is a **depth-first walk of the decision tree**: take each decision **False first**; to advance, **flip the deepest decision that still has an untried value**; when a decision's *both* values are done under the current prefix, **backtrack up** to the previous decision and flip that, then resume **false-first** for everything newly reached. Example (decisions at lines 2, 3-nested-in-2, and 5):

```
1 int g(int p,int q){
```



```

2  if (p>0)                // decision @2
3      if (q>0) return 1;  // decision @3  (reached only when @2 is True)
5  if (p==q) return 2;     // decision @5
6  return 3; }

```

- **Path 1** — all False: @2 (F) skips the block (so @3 isn't reached), @5 (F) → 1, 2 (F), 5 (F), 6.
 - **Path 2** — flip the last decision (@5) → True: 1, 2 (F), 5 (T).
 - **Path 3** — under @2 (F), @5 is now done both ways ⇒ **backtrack to @2, flip to True**; @3 is reached for the first time, taken **False first**, then @5 False: 1, 2 (T), 3 (F), 5 (F), 6.
 - (next: 1, 2 (T), 3 (F), 5 (T), then backtrack to @3 → 1, 2 (T), 3 (T)....) A **nested if** only enters the tree once its **enclosing** decision is True — that's why @3 appears only after @2 flips to True.
- **for loop — the update (i++) runs *last* in each iteration.** For `for (i=0; i<n; i++) { body }` the order per iteration is **init** → **test i<n** → **body** → **i++** → **back to test**. So in the PV/PC table the i++ row comes **after** the whole body, not next to the i<n test — a common ordering slip when a use of i inside the body must see the *pre-increment* value.
 - Assignments update PV only; branches update PC only — never both on one row.
 - **"Does symbolic execution guarantee full branch coverage?"** (recurring true/false — state assumptions explicitly, then split into three cases; a tiny example for each):
 1. **Infeasible branch** → **doesn't count**. Symbolic execution can't find an input for it (its PC is UNSAT), **but no test suite could cover it either**, so it's not a real gap. *Example:* `if (x > 10) { if (x < 5) DEAD; }` — reaching DEAD needs $x > 10 \wedge x < 5$, which is UNSAT; that edge is uncoverable by *anyone*, so failing to cover it isn't a failure of symbolic execution.
 2. **Loops / unbounded (or too-large) tree** → **may not terminate**. The tree can be infinite, so the run might never finish. *Example:* `while (i < n) i++;` with n symbolic unfolds to 0, 1, 2, ... iterations — infinitely many paths. If the question *assumes* "we run symbolic execution" means it **does** finish exploring the whole tree, the statement is **trivially true**; otherwise it may cover nothing conclusive.
 3. **Otherwise (bounded tree, fully explored)** → **true, and stronger**. *Example:* `if (x>0) A else B; if (y>0) C else D;` — no loops, exactly 4 feasible paths; symbolic execution walks all 4, so it takes **both** edges of each `if` (full branch) **and** every path (full **path** coverage $\supset$ branch coverage). ⇒ Under the "whole tree explored" assumption, symbolic execution gives **full branch coverage** (in fact full *path* coverage); the only escapes are non-termination (case 2) or genuinely infeasible — hence uncoverable — branches (case 1).

Cheat sheet.

- Columns `line` | `PV` | `PC`. Conjunction `/\`, negation `!/¬`. Inputs UPPERCASE; array length `SIZE_OF_A`.
- **False branch first**; negate-last-constraint to explore sibling; simplify & state equivalences; always write explicit symbolic return + full PC; give satisfying input only when SAT.
- Branch denominator = $2 \times \text{\#decisions}$.

7. Concolic Testing (DART & CUTE)

Plain words

Pure symbolic execution (§6) breaks down when the maths gets too hard for the solver — a non-linear formula, a function whose source you don't have, a messy pointer. Concolic testing fixes this by running the program on **real inputs and symbols at the same time** ("concrete + symbolic" = concolic). It keeps the symbolic PC to reason about paths, but whenever the solver gets stuck it just plugs in the *actual concrete value* from the real run and moves on. To reach a new path it takes the last branch condition and flips it, then asks the solver for an input satisfying the flipped condition — repeat until you hit the target (e.g. ERROR).

Key definitions.

- **Concolic = concrete + symbolic, side by side.** The real (concrete) values keep the program running; the symbolic side reasons about paths. When the solver can't cope (an opaque/non-linear function, a pointer), you **fall back to the concrete value** instead of getting stuck.
- **DART (Directed Automated Random Testing)** — start from random inputs, record the branch conditions hit, then negate them one at a time to steer execution down not-yet-taken paths. For a black-box function it just substitutes the concrete number the function actually returned.
- **CUTE (Concolic Unit Testing Engine)** — extends this to pointers and dynamic data structures (linked lists, trees) using **logical addresses**: rather than reasoning about raw memory addresses (which change run to run), it treats "same value $\Rightarrow$ same logical location". NULL tests become symbolic constraints like `P==NULL`, `PN==NULL`.
- **Pointer symbol notation** (how a pointer chain maps to symbols): `p→P`, `p→v→PV` (the value field), `p→next→PN` (the next pointer), `p→next→v→PNV`, `p→next→next→PNN`, ... — i.e. append a letter per field you follow.

The recipe — columns `line` | `concrete state` | `PV (symbolic)` | `PC`:

1. Pick initial input —

- **Linked-list / int-from-zero**: first random int starts at **0**, increment by 1 until the PC holds; pointers start `NULL`.

- Usually you are told which initial values to choose.

2. **Run the table:** concrete column = real values / data-structure graph; PV = symbols; PC appends each branch constraint with $\wedge$, mark (True) / (False). Black-box result = $f(x)$: PV gets token THIRD_PARTY_FUNCTION; concrete column gets the *actual computed number*.
3. **Report:** concrete input, concrete output, symbolic PC.
4. **Negate the last branch constraint**, solve for next input (increment int from 0 until PC holds; grow list by one cell when $\rightarrow_{next} \neq \text{NULL}$ needed).
5. **Repeat until ERROR.** Linked-list NULL-check $\Rightarrow$ **4 iterations**; black-box equality $\Rightarrow$ **2 tables**.

Worked example A — CUTE, linked list, start ints at 0 (full 4-iteration walk-through).

```
void bar(cell* p){
1:  if (p == NULL || p->next == NULL) return;
2:  if (p->v > p->next->v) ERROR;
}
```

Pointer symbols: $p \rightarrow P$, $p \rightarrow v \rightarrow PV$, $p \rightarrow \text{next} \rightarrow PN$, $p \rightarrow \text{next} \rightarrow v \rightarrow PNV$. Each iteration: pick the input, run the table, read off output, then **negate the last branch constraint** to derive the next input.

Iteration 1 — Input: $p = \text{NULL}$ (pointers start NULL).

line	concrete state	PV (symbolic)	PC
entry	$p = \text{NULL}$	$p \rightarrow P$	—
1	$p == \text{NULL} \rightarrow \text{true}$ (short-circuits, 2nd atom not evaluated)	—	$P == \text{NULL}$ (True) $\rightarrow \text{return}$

Output: **returns normally, no ERROR.** **Negate** $P == \text{NULL}$ (True) $\rightarrow P \neq \text{NULL} \Rightarrow$ next input needs a non-null pointer $\rightarrow$ build a **1-cell** list.

Iteration 2 — Input: $p = [v=0] \rightarrow \text{NULL}$ (1 cell).

line	concrete state	PV (symbolic)	PC
entry	$p = [0] \rightarrow \text{NULL}$	$p \rightarrow P$, $p \rightarrow v \rightarrow PV$, $p \rightarrow \text{next} \rightarrow PN$	$P \neq \text{NULL}$ (carried from the flip)
1	$p == \text{NULL}$ false; $p \rightarrow \text{next} == \text{NULL}$ true $\rightarrow$ cond true	—	$P \neq \text{NULL} \wedge PN == \text{NULL}$ (True) $\rightarrow \text{return}$

Output: **returns normally, no ERROR.** Negate $PN == NULL$ (True) $\rightarrow PN != NULL \Rightarrow$ next input needs a non-null $next \rightarrow$ grow to a **2-cell** list.

Iteration 3 — Input: $p = [v=0] \rightarrow [v=0] \rightarrow NULL$ (2 cells, values from 0).

line	concrete state	PV (symbolic)	PC
entry	$p = [0] \rightarrow [0] \rightarrow NULL$	..., $p \rightarrow next \rightarrow v \rightarrow PNV$	$P != NULL \wedge PN != NULL$ (carried)
1	false false $\rightarrow$ fall through	—	$P != NULL \wedge PN != NULL$ (False)
2	$p \rightarrow v > p \rightarrow next \rightarrow v \rightarrow 0 > 0$ false $\rightarrow$ no ERROR	—	... $\wedge PV > PNV$ (False) $\rightarrow$ return

Output: **returns normally, no ERROR.** Negate $PV > PNV$ (False) $\rightarrow PV > PNV$ (True) $\Rightarrow$ need the first value bigger: increment to $PV=1$, $PNV=0$.

Iteration 4 — Input: $p = [v=1] \rightarrow [v=0] \rightarrow NULL$.

line	concrete state	PV (symbolic)	PC
entry	$p = [1] \rightarrow [0] \rightarrow NULL$	as above	$P != NULL \wedge PN != NULL$ (carried)
1	false false $\rightarrow$ fall through	—	$P != NULL \wedge PN != NULL$ (False)
2	$1 > 0$ true $\rightarrow$ ERROR	—	$P != NULL \wedge PN != NULL \wedge PV > PNV$ (True) $\rightarrow$ ERROR

Output: **ERROR reached.** Final PC: $P != NULL \wedge PN != NULL \wedge PV > PNV$; input that triggers it: the 2-cell list $[1] \rightarrow [0]$.

Worked example B — black-box `thirdPartyFunction`, start $x=y=1$ (full 2-table walk-through).

```
computeResult(x, y){
1:  result = thirdPartyFunction(x);    // hidden: f(x) = 600x3 + 700x2 + 900x + 8
2:  if (result == y) ERROR;
3:  return result;
}
```

Symbols $x \rightarrow X$, $y \rightarrow Y$; `result` gets the opaque token `THIRD_PARTY_FUNCTION` because the solver can't see inside `f`.

Table 1 — Input: $x = 1, y = 1$.

line	concrete state	PV (symbolic)	PC
entry	$x=1, y=1$	$x \rightarrow X, y \rightarrow Y$	—
1	engine runs $f(1)=2208 \rightarrow$ $result=2208$	$result \rightarrow \text{THIRD_PARTY_FUNCTION}$	—
2	$2208 == 1$ false $\rightarrow$ no ERROR	—	$\text{THIRD_PARTY_FUNCTION} != Y$ (False)
3	return 2208	—	—

Output: **returns 2208, no ERROR.** Negate $\text{THIRD_PARTY_FUNCTION} != Y \rightarrow$

$\text{THIRD_PARTY_FUNCTION} == Y$. The solver can't invert f , so **reuse the concrete output**: keep $x=1$ (so $result$ stays 2208) and set $y = 2208$.

Table 2 — Input: $x = 1, y = 2208$.

line	concrete state	PV (symbolic)	PC
entry	$x=1, y=2208$	$x \rightarrow X, y \rightarrow Y$	—
1	$f(1)=2208 \rightarrow$ $result=2208$	$result \rightarrow \text{THIRD_PARTY_FUNCTION}$	—
2	$2208 == 2208$ true $\rightarrow$ ERROR	—	$\text{THIRD_PARTY_FUNCTION} == Y$ (True) $\rightarrow$ ERROR

Output: **ERROR reached.** Input that triggers it: $(x=1, y=2208)$.

Exam patterns & gotchas.

- **Basic-condition denominator** = $2 \times \# \text{atomic conditions}$. `bar` has 3 atoms $\Rightarrow$ 6.
- "Can you get full branch coverage *without* full basic-condition coverage?" (a common yes/no sub-question) — **yes**, because of short-circuiting. In `if (p==NULL || p->next==NULL)`: a test with `p==NULL` takes the true branch (and `p->next==NULL` is never even evaluated, thanks to `||` stopping early); a test with a full 2-cell list takes the false branch. Both branches are now covered, yet the atom `p->next==NULL` was never made true on its own $\Rightarrow$ basic-condition coverage is still incomplete.
- **Iterations**: NULL-check list $\Rightarrow$ 4; black-box equality $\Rightarrow$ 2 tables. (4 is the suggested count; other counts can be acceptable.)
- **Black-box/non-linear**: keep a symbolic token, fill concrete column with the real value, and reuse that concrete value as the next input when inverting. Never algebraically

solve the opaque function.

- **Choose the initial input by convention:** 0-and-increment for lists, 1 for black-box; pointers start NULL.
- Always **negate the last constraint** (not an earlier one); grow the structure by one node when the negated constraint needs a non-null `next`.
- **Negation is per *atomic predicate actually evaluated*, never the whole source `&&/||`.** Concolic instruments individual conditional statements, and short-circuit means an un-evaluated operand is **not** in the PC. So for `if (p==NULL || p->next==NULL)` with `p=NULL`: only `P==NULL` (true) is recorded (the `||` stops early, so `p->next==NULL` isn't evaluated) — the next run negates **just** `P==NULL` $\rightarrow$ `P!=NULL`, not the whole condition. (This is why `bar()` needs 4 iterations: the atoms `P==NULL`, then `PN==NULL`, then `PV>PNV` are peeled one run at a time. Refs: DART & CUTE papers — path constraint = conjunction of executed predicates, negate one conjunct.)

Cheat sheet.

- Columns `line | concrete | PV | PC. /\ conj, || disj, mark (True)/(False)`.
- Pointer symbols `P, PV, PN, PNV, PNN, ...`; black-box token `THIRD_PARTY_FUNCTION`.
- Per iteration write: concrete input, concrete output, symbolic PC, how next input is chosen.
- Random ints start **0** (lists) or **1** (black-box); pointers NULL; iterate negate $\rightarrow$ resolve until ERROR.
- Basic-condition denominator = $2 \times \text{\#atomic conditions}$.

8. FSM-based Testing (UIO, DS, W-set)

Plain words

Some systems have *memory* — the same input does different things depending on what happened before (a vending machine, a login flow). We model these as a **Finite State Machine (FSM)**: a set of states with labelled transitions ("on input `a`, go from state `s0` to `s1` and output `0`"). To test such a system you need to confirm it's really *in* the state you think it is. The three tools all answer "which state am I in?" by feeding inputs and watching outputs: a **UIO** is a fingerprint for *one* state, a **DS** is a single fingerprint that identifies *every* state at once, and a **W-set** is a *collection* of short inputs that together tell all states apart. UIO and DS don't always exist; a W-set always does (for a well-behaved FSM).

Key definitions.

- **Mealy FSM** = $\langle S, I, O, s_0, \delta, \lambda \rangle$ — states `S`, inputs `I`, outputs `O`, start state `s0`, a next-state function δ (state+input $\rightarrow$ state), an output function λ (state+input $\rightarrow$ output). "Mealy" means the output is produced **on the transition** (it depends on both the current state and the input), not just on the state.

- **Four assumed properties: completely specified** (every state has a defined transition & output for every input — δ, λ are "total"), **deterministic** (one input $\rightarrow$ exactly one next state), **reduced** (no two states behave identically — otherwise you couldn't tell them apart), **strongly connected** (you can get from any state to any other). The UIO/DS/W theory assumes *reduced*.
- **UIO (Unique Input-Output) for state s_i** — an input sequence whose *output* is **different from what every other state would produce** on that same sequence. So observing that output proves "I was in s_i " — a fingerprint for one state. An FSM "has a UIO" only if **every** state has one.
- **Distinguishing Sequence (DS)** — one *single* input sequence that yields a **different output for every state** — one fingerprint that identifies all states at once. A DS $\Rightarrow$ every state trivially has a UIO. Not every reduced FSM has a DS.
- **Characterizing set W** — a *set* of sequences $\{w_1, \dots, w_k\}$ that **together** tell all states apart (no single one has to; the combination does). Always exists for a reduced FSM. A DS is just the special case where one sequence suffices ($|W|=1$).
- Key implication (exam favorite): **if even one state has no UIO $\Rightarrow$ there is no DS** (because a DS would hand every state a UIO). The reverse isn't true.

Small example — one FSM, all three fingerprints side by side. 3 states, $I=\{a,b\}$, $O=\{0,1\}$ (read $s_2 / 0$ as "go to s_2 , output 0"):

state	on a	on b
s1	$s_2 / 0$	$s_1 / 0$
s2	$s_3 / 1$	$s_1 / 0$
s3	$s_1 / 1$	$s_2 / 0$

- **DS = aa .** Feed the input sequence aa starting from each state and read off the two outputs: $s_1 \rightarrow 01$, $s_2 \rightarrow 11$, $s_3 \rightarrow 10$. All three output strings differ $\Rightarrow$ this one sequence identifies *every* state. (Trace s_1 : a outputs 0 and moves $s_1 \rightarrow s_2$, then a outputs 1 and moves $s_2 \rightarrow s_3 \Rightarrow 01$.)
- **UIO per state** (one fingerprint each; lengths may differ). Length-1 a gives outputs $s_1=0$, $s_2=1$, $s_3=1$ — so output 0 on a is unique to **$s_1 \Rightarrow \text{UIO}(s_1) = a$** . s_2 and s_3 tie on a (both 1) and b outputs 0 everywhere, so they need length 2: **$\text{UIO}(s_2) = aa$** (output 11), **$\text{UIO}(s_3) = aa$** (output 10). Note a DS automatically serves as a UIO for every state — that's why aa works for all three.
- **$W = \{aa\}$.** Because a DS exists, the characterizing set collapses to just that one sequence ($|W|=1 \Leftrightarrow$ the single word *is* a DS). No length-1 set could do it here — a can't separate s_2 from s_3 and b outputs 0 for all states, so s_2 and s_3 only diverge from length 2 onward. Contrast **Worked example 3** below, where there is *no* DS and W genuinely needs two words $\{a, b\}$.

The recipes.

(a) Find or refute a UIO for a state s_i — the **UIO tree**. Goal: find the shortest input sequence w whose output from s_i differs from the output *every other* state gives on that same w . Observing that output then proves "I was in s_i ." You search for w by growing a tree one input at a time.

The idea in one sentence: **start by assuming every state could be mistaken for s_i , then feed inputs that peel away the states whose output differs, until only s_i is left.** The group you track is the **look-alike set** — the states that, on the inputs applied so far, have produced the **exact same output string as s_i** (so from the outside they still look identical to s_i , i.e. you cannot yet tell them apart from it). For each look-alike also record **which state it has now moved to**, since that determines its next output.

- **Root (no input yet):** no output seen, so no state can be ruled out — the look-alike set is **all states**, each still sitting at itself.
- **Apply input x :** compute the output x gives from s_i 's *current* state — call it o . Any look-alike whose output on $x \neq o$ has just revealed itself as different $\Rightarrow$ **remove it**. The ones whose output $= o$ still look like s_i ; advance each of them (and s_i) to its next state.
- **Success:** the look-alike set shrinks to just $\{s_i\}$ — no other state still matches s_i 's output string $\Rightarrow$ the inputs along this path are a **UIO** for s_i .
- **Dead branch:** s_i 's current state becomes the **same** state as another look-alike's current state (a **collision** — from here they give identical output and next-state forever, so no input can ever separate them), or the look-alike set repeats one seen earlier (a loop).
- **No UIO:** if *every* branch dies (collision/loop) before the set reaches $\{s_i\}$, then s_i has no UIO.

Worked mini-example (the 3-state machine above): find UIO(s_2). "look-alikes" = states still matching s_2 's output so far; @ shows where each has moved.

```
start: look-alikes {s1, s2, s3}           (no input yet — anyone could be s2)
      └ a → s2 outputs 1 (moves →s3). s1 outputs 0 ≠ 1 ⇒ removed. s3 outputs 1 = 1 ⇒
          look-alikes { s2@s3 , s3@s1 }    (s3 still matches s2's output "1" so far)
          └ a → s2 (now at s3) outputs 1 (→s1). s3 (now at s1) outputs 0 ≠ 1 ⇒ remove
              look-alikes { s2 }           ★ only s2 remains ⇒ UIO(s2) = `aa` (s2's
```

UIOs really can be a *different word per state* (each state's tree terminates as soon as *that* state is isolated — via whatever input does it, at whatever depth). Worked example:

state	on a	on b
s1	s1 / 1	s2 / 0
s2	s3 / 0	s2 / 1
s3	s2 / 0	s1 / 0

- **UIO(s1) = a** — a gives s1 output 1, unique (s2,s3 give 0). Tree isolates s1 in **one** step via a.
- **UIO(s2) = b** — b gives s2 output 1, unique (s1,s3 give 0). Isolated in one step via a **different input** (b, not a).
- **UIO(s3) = ab** — s3 outputs 0 on both a and b (ties with someone each time), so length-1 fails; ab gives s3 → 01, unique (s1→10, s2→00). Needs **two** steps.

So the three UIO trees genuinely diverge: a, b, ab — different first inputs *and* different lengths. (This machine also happens to have a DS = ab, which is why s3's shortest UIO equals it — but s1 and s2 get away with far shorter, state-specific words. That's the whole point of UIOs over a DS: **per-state, often cheaper.**)

Reading it: the first a already peels off s1 (it alone output 0), but s3 still shadows s2 (both output 1); the second a finally splits them (s2 → 1, s3 → 0), leaving s2 alone. s1 needed no tree at all — a single a makes its output 0 unique (see the small example above). **To refute** a UIO you run the same tree and show every branch hits a collision or loop — worked next.

(b) *DS tree*: node = **partition of S into blocks**. Develop on input x: within each block group states by output on x; child blocks = each group's next-states. Prune:

Rule	Condition	Meaning
D1 homogeneous	a block has a repeated state	inseparable → prune (dead)
D2 singleton	every block is a singleton	root→node path is a DS (success)
D3 loop	child block already on root→node path	prune

First D2 ⇒ DS = that input path. All branches die D1/D3 ⇒ **no DS**.

(c) *Characterizing set W*: build the output table for short words (length 1, then 2, ...); greedily pick words so **every pair of states differs on ≥1 word**; present W + per-state output table; the per-state output **column-vectors must all be distinct (no 2 columns have the same values)**.

(d) *Conformance tests*: "conformance testing" = checking a real implementation matches the FSM spec. A **transfer sequence** $\text{transfer}(s_i)$ is just a shortest input sequence that drives the machine from the start state s_0 to state s_i (so you can reach the state you want to test). Build them with a BFS **spanning tree** from s_0 ; the collection is the **state cover**. Then v is your chosen state-identifier (UIO, DS, or W).

- **State coverage** (verify every state exists): for each s_i , run $\text{transfer}(s_i) \cdot v(s_i)$ — go to s_i , then apply its fingerprint to confirm you're really there.
- **Transition coverage** (verify every transition, stronger): for each edge $s_i \xrightarrow{x} s_j$, run

$\text{transfer}(s_i) \cdot x \cdot V(s_j)$ — go to s_i , take input x , then fingerprint to confirm you landed in the expected s_j . Transition coverage $\supseteq$ state coverage ($\supseteq$ = strictly stronger). (Weaker alternative: a **transition tour** from s_0 that just walks every edge and checks outputs, without confirming the target state.)

Worked example 1 — prove no UIO $\Rightarrow$ no DS. 3 states, $I=\{a,b\}$, $O=\{0,1\}$:

state	a	b
s0	s1 / 0	s2 / 0
s1	s0 / 1	s2 / 0
s2	s1 / 0	s0 / 1

Why s0 has no UIO — run the UIO tree from s0 and try each possible *first* input:

- Start with **a**: s0 outputs 0 and goes to s1 — but s2 *also* outputs 0 and *also* goes to s1. After a, s0 and s2 sit in the **same state** having produced the **same output**, so from here they behave identically forever. **Collision** — this branch is dead.
- Start with **b**: s0 outputs 0 and goes to s2 — but s1 *also* outputs 0 and goes to s2. Same trap, this time with s1. Dead.

Both possible first inputs trap s0 in a collision (same output **and** same next-state as another state), and a collision can never be undone $\Rightarrow$ **s0 has no UIO**. And a single state with no UIO is enough to conclude **there is no DS** (a DS would have to hand *every* state a UIO). (*The DS tree agrees: both children of the root contain a repeated state — rule D1.*)

If instead a question changes one edge — say $s_2 \xrightarrow{b} s_0$ becomes $s_2 \xrightarrow{b} s_0 / 0$: first re-check the changed edge, but the two blocking collisions above (on a: $s_0, s_2 \rightarrow s_1/0$; on b: $s_0, s_1 \rightarrow s_2/0$) don't involve it $\Rightarrow$ **still no UIO, no DS**. (Always re-check the changed edge first.)

Worked example 2 — DS exists; show the tree. 5 states; following branch **aba**:

```

root      [ {0,1,2,3,4} ]
└─a→      [ {0,2,3}, {1,2} ]
          └─b→ [ {0},{0},{3,4},{3} ]
                └─a→ [ {1},{2},{2},{3},{3} ]    ★ D2 all singletons  $\Rightarrow$  DS = a·b·a

```

Verification (all 5 outputs of aba distinct): $s_0=001$, $s_1=100$, $s_2=101$, $s_3=110$, $s_4=010$.

(b) Minimum DS length for n states. A DS must give each of the n states a **distinct output string**. A length- L sequence over an output alphabet of size m can produce only m^L different strings, so to tell n states apart you need $m^L \geq n \Rightarrow L \geq \lceil \log_m n \rceil$. Just take the **smallest L with $m^L \geq n$** (unambiguous; equals $\lceil \log_m n \rceil + 1$ for non-power n — same value, so the course writes it " $\log_m n + 1$ " — but do **NOT** compute $\lceil \log_m n \rceil + 1$). Examples with $m=2$: $n=5 \rightarrow 2^L \geq 5 \Rightarrow L=3$ ($2^2=4 < 5 \leq 8=2^3$); $n=23 \rightarrow 2^L \geq 23 \Rightarrow L=5$ ($2^4=16 < 23 \leq 32=2^5$).

Worked example 3 — no DS, give W. FSM: $s_0 \xrightarrow{a/0} s_1, s_0 \xrightarrow{b/0} s_2; s_1 \xrightarrow{a/0} s_1, s_1 \xrightarrow{b/1} s_1; s_2 \xrightarrow{a/1} s_2, s_2 \xrightarrow{b/0} s_2$. Root: on **a**, s_0, s_1 both $\rightarrow s_1/0$ (D1); on **b**, s_0, s_2 both $\rightarrow s_2/0$ (D1) $\Rightarrow$ **no DS**. **W = {a, b}** (minimal): *a* separates $\{s_0, s_1\}$ from s_2 ; *b* separates $\{s_0, s_2\}$ from s_1 . Output vectors $s_0=(0,0), s_1=(0,1), s_2=(1,0)$ — distinct ✓. Dropping either word merges a pair.

Worked example 4 — full conformance test table ("write all the test cases").

state	a	b
s0	s1 / 0	s3 / 1
s1	s2 / 1	s2 / 0
s2	s0 / 1	s2 / 1
s3	s2 / 1	s1 / 1

This FSM has **no single DS**, so the **state-verification sequence is the reached state's UIO** — $UIO(s_0)=a, UIO(s_1)=b, UIO(s_2)=aa, UIO(s_3)=bb$. **Reset = R** (applied last). **Transfer sequences** from s_0 : $transfer(s_0)=\epsilon, transfer(s_1)=a, transfer(s_2)=aa, transfer(s_3)=b$. The *Input sequence* column is the whole run — **transfer @ input-under-test @ UIO(reached) @ R** — with @ separating the four parts. There are 4 states $\times$ 2 inputs = 8 transitions $\Rightarrow$ 8 rows:

State	Input under test	Input sequence	Expected output (input)	Reached	Expected output (verification UIO)
s0	a	a@b@R	0	s1	0
s0	b	b@bb@R	1	s3	10
s1	a	a@a@aa@R	1	s2	10
s1	b	a@b@aa@R	0	s2	10
s2	a	aa@a@a@R	1	s0	0
s2	b	aa@b@aa@R	1	s2	10
s3	a	b@a@aa@R	1	s2	10
s3	b	b@b@b@R	1	s1	0

Reading a row (s_2, a): reset to s_0 , run $transfer(s_2)=aa$ to reach s_2 , apply the input-under-test *a* (expect output **1**, lands in **s0**), then apply $UIO(s_0)=a$ and expect its output **0** — which confirms

you really reached s_0 ; finally R resets. Columns 5–6 come straight from the table: column 5 = $\delta(\text{state}, \text{input})$, column 6 = the reached state's own UIO output. (*Because there's no DS here, the verification sequence **changes per row** — it's whichever UIO belongs to the reached state.*)

Exam patterns & gotchas.

- **Prove non-existence rigorously:** give the structural reason — the (output, next-state) **collision** between two states means no input ever separates them; back it with the pruned tree (all branches D1/D3).
- **No UIO $\Rightarrow$ no DS** (use freely); reverse is false.
- **No DS $\Rightarrow$ use W** (always works for reduced FSM); substitute W wherever you'd use the DS in conformance tests.
- **"Change one label":** re-check whether the changed edge is one of the colliding ones; one label change can create or destroy a DS.
- **Which input is the "input under test" in a state-verification row?** To verify that transition $s_i \xrightarrow{x} s_j$ lands you in s_j , the **input under test is x** (the transition's own input) — you apply x , then apply s_j 's **DS/UIO** to confirm you actually reached s_j .

Cheat sheet — UIO vs DS vs W:

	UIO	DS	W
What	one seq per state	one seq for all states	a <i>set</i> of seqs
Count	n seqs (varied lengths)	1	$k \geq 1$
Always exists?	No	No	Yes (reduced FSM)
Built via	UIO tree	DS tree	output table, pair-cover
Relation	DS $\Rightarrow$ all UIOs	DS $\Rightarrow$ each UIO	$\text{len}(W) = 1 \Rightarrow$ that word is a DS

DS-tree pruning: **D1** repeated state in a block = dead; **D2** all singletons = DS found; **D3** repeated block = loop. Conformance: spanning tree $\rightarrow$ state cover $\rightarrow \text{transfer} \cdot V$ per state; transition cover = $\text{transfer}(s_i) \cdot x \cdot V(s_j)$ per edge.

9. Black-box Techniques (ECP, BVA, Decision Tables, Domain)

Plain words

"Black-box" means you pick test inputs from the *specification* alone, without looking at the code inside. The problem is still "too many inputs" — so these techniques are smart ways to choose a few representatives. **ECP**: group inputs that *should be treated the same* and test one from each group. **BVA**: bugs love edges, so test right at and just past the boundaries between groups.

Decision tables: when the output depends on several yes/no conditions, tabulate the combinations. **Domain testing**: picture the input space as regions separated by boundary lines, and test points *on and just off* each boundary to catch a mis-drawn boundary.

Black-box has four techniques. Each is given below as **definition-in-context** → **recipe** → **worked example** (there is no separate glossary — every term is defined the first time the technique that needs it uses it). Running spec for the examples: $\text{discount}(\text{qty}) = 0\%$ if $\text{qty} < 10$, 10% if $10 \leq \text{qty} \leq 99$, 20% if $\text{qty} \geq 100$; qty is a positive int.

① **Equivalence Class Partitioning (ECP)**. An **equivalence class (EC)** is a group of inputs the program *should* treat identically (e.g. "all ages 18–65"); you split the input space into **valid** classes and **invalid** classes, and test **one representative per class** — that one input stands in for the whole class, which is the saving.

- *Recipe*. A **range** → 1 valid + 2 invalid (one below, one above); a **set/enum** → 1 valid per member + 1 invalid (a non-member); a **"must be X"** rule → 1 valid + 1 invalid. Combine valid classes into shared tests, but give **each invalid class its own test** — never combine two invalids (below).
- *Worked example*. $\text{discount}(\text{qty})$: the one invalid class is $\text{qty} \leq 0$; the valid classes are $[1, 9]$, $[10, 99]$, $[100, \infty)$. Four tests, e.g. $\text{qty} = -3$ (invalid), 5, 50, 500.

② **Boundary Value Analysis (BVA)**. A **boundary value** is an input right at, or one step past, the edge of an EC. Off-by-one and $<$ -vs- $\leq$ bugs cluster exactly here, so these are the highest-value tests. BVA **extends** ECP (it adds edge tests to the class representatives), it doesn't replace it.

- *Recipe*. For a class spanning $[a, b]$, test $a - \epsilon$, a , $a + \epsilon$ and $b - \epsilon$, b , $b + \epsilon$ plus one nominal interior value ($\epsilon =$ one smallest step; $\epsilon = 1$ for ints). Classic $[-10, 10] \rightarrow -10.1, -10, -9.9, 9.9, 10, 10.1$.
- *Worked example*. $\text{discount}(\text{qty})$ boundaries are $9|10$ and $99|100$, so test 0, 1, 9, 10, 99, 100, large. The pairs straddling $9|10$ and $99|100$ are what expose a closure or shift fault in the $</\leq$ operators.

③ **Decision Tables**. A **decision table** lists **conditions** down the side (each cell **Y / N / –**, where – = "doesn't matter") against **rules** in the columns, mapping each rule to its **action(s)**. With n conditions there are up to 2^n rule columns; you **merge** columns that produce the same action using – don't-cares, and each surviving column becomes one test case. (*Its systematic cousin, category-partition, works spec* → **categories** (input characteristics) → **choices** (the partitions of each) → **constraints** that prune nonsense combos ($[property]$, $[if...]$, $[error]$, $[single]$) → **test frames** = concrete combinations.)

- *Recipe*. Enumerate the 2^n rules → fill in each rule's action → merge adjacent columns

with identical actions into a – column → one test per surviving column.

- *Worked example.* `shipping(member, orderOver50) = free` if `member` OR `order ≥ $50`, else `$5`. Two conditions ⇒ $2^2=4$ rules; three collapse into one via don't-cares:

Condition	Rule A	Rule B	Rule C
member?	Y	N	N
order ≥ \$50?	–	Y	N
→ action	free	free	\$5

Rule A merges the two original `member=Y` columns (shipping is free regardless of order size). Three columns ⇒ **3 tests**: (`member, any`), (`non-member, $60`), (`non-member, $20`).

④ **Domain Testing.** Picture the program as a **classifier**: it sorts each input into a **subdomain** (region of the input space) and computes a result for that region. Two fault kinds follow: a **domain error** = the input landed in the *wrong region* because a **predicate** (`if` condition) is wrong; a **computation error** = right region but *wrong value* because an **assignment** is wrong. Domain testing hunts the **domain (boundary) errors**. Supporting vocabulary, defined where it bites:

- *Boundary geometry.* A **closed** boundary *includes* its edge points ($\leq / \geq$); an **open** boundary *excludes* them ($< / >$). **Adjacent domains** share a boundary; an **extreme point** is where two boundaries cross.
- *Three ways a boundary is mis-coded.* **Closure** ($\leq$ written as $< —$ right line, wrong include/exclude), **shifted** (right slope, wrong constant, e.g. $x+y>5$ coded $x+y>4$), **tilted** (wrong coefficient, e.g. $x+y>5$ coded $x+0.5y>5$).
- *ON / OFF points.* An **ON point** lies *exactly on* the boundary (the equality holds). An **OFF point** lies *just off* it — and which side flips with closure: for a **closed** boundary the OFF sits just *outside* (in the adjacent domain), for an **open** boundary it sits just *inside*. (This flip is the single biggest point-earner — see gotchas.)
- *Recipe (ON–OFF–ON).* Per boundary, pick two **ON** points A, B spread apart along the boundary line + one **OFF** point C → test the sequence A, C, B. The two ONs catch a **tilt** (a wrong slope shows at one end), the ON/OFF pair catches **shift + closure**.
- *Worked example.* `discount(qty)` boundary `qty=10` is **closed on the 10%-side** ($10 \leq qty$). ON = 10 (should give 10%), OFF = 9 (just outside → 0%). If the code wrongly wrote `qty>10` (a **closure** error), the ON point 10 falls through to 0% instead of 10% ⇒ the test catches it.

Exam patterns & gotchas.

- *Never merge two invalid ECs* — can't tell which triggered the failure.
- Domain error = conditional fault; computation error = assignment fault. Boundary-interior

/ ON-OFF target **domain (boundary) errors**.

- **Closed boundary** → **OFF outside; open boundary** → **OFF inside** (#1 deduction).
- BVA *extends* ECP, doesn't replace it.

Cheat sheet.

Concept	One-liner
ECP valid / invalid	1 per class, combine / 1 EACH, never combine
BVA range [a,b]	$a-\epsilon$, a , $a+\epsilon$, $b-\epsilon$, b , $b+\epsilon$, nominal
Decision table	rules= 2^n , merge don't-cares, 1 col = 1 test
Computation error	correct path, wrong value (assignment)
Domain error	wrong path, faulty predicate
Closure / shifted / tilted	$\leq \leftrightarrow <$ / wrong constant / wrong coefficient
ON / OFF (closed) / OFF (open)	on boundary / just outside / just inside
Criterion	ON–OFF–ON (A, C, B) per boundary

10. JUnit & Tooling Reference (Pitest, JaCoCo)

Plain words

This section is the *practical* toolkit — the actual Java tools that implement the ideas above. **JUnit** = the framework you write tests in (an `assert...` that throws if the program misbehaves). **Pitest** = the tool that automates mutation testing from §1 (it plants the mutants and runs your tests against each). **JaCoCo** = the tool that measures coverage from §2 (which lines/branches your tests actually ran). Exam questions here are usually "given this code and these tests, what does the tool report?" — so know how each tool *counts*.

Key definitions.

- **JUnit**: the Java unit-test framework. A test does: create the object → set up inputs → state the expected result → run the code → **assert** the result matches. A failed assertion throws `AssertionFailedError`; related tests group into *suites*.
- **Mutant** / **killed** / **survived** / **equivalent** and the **score** $100 \cdot D / (N - E)$ — all defined in §1.
- **Pitest (PIT)**: a JVM mutation-testing tool. It applies its default **mutators** (the tiny changes) to your compiled bytecode and re-runs your test suite once per mutant, reporting which survived.

- **JaCoCo**: a coverage tool. It has several counters (instruction / line / branch / complexity / method / class); the exam cares about **statement (line)** and **branch** coverage. Run with `mvn test jacoco:report`.

JUnit assertions (write precisely).

Assertion	Use
<code>assertEquals(expected, actual)</code>	object/primitive equality (<code>.equals</code>)
<code>assertEquals(expected, actual, delta)</code>	doubles/floats — MUST give a tolerance (e.g. <code>1e-9</code>)
<code>assertTrue(cond)</code> / <code>assertFalse(cond)</code>	booleans
<code>assertNull</code> / <code>assertNotNull</code>	null checks
<code>assertSame</code> / <code>assertNotSame</code>	reference identity (==) , not value
<code>assertArrayEquals(exp, act)</code>	array contents
<code>assertThrows(Ex.class, exec)</code>	code throws the expected exception (returns it to inspect)
<code>fail(msg)</code>	force failure (unreached-branch guards)

JUnit example (one method + one test). Minimal shape — a test is **Arrange** → **Act** → **Assert**; `@Test` marks it, a failed assertion throws:

```
static int abs(int x) { return x < 0 ? -x : x; }    // method under test
```

```
import org.junit.jupiter.api.Test;
import static org.junit.jupiter.api.Assertions.*;

class AbsTest {
    @Test
    void abs_ofNegative_isPositive() {
        int x = -3;           // Arrange
        int r = abs(x);       // Act
        assertEquals(3, r);   // Assert
    }
}
```

Scale up by adding **one @Test per behaviour** (a boundary like `abs(0)`, an error path via

`assertThrows`, a **delta on double** `assertEquals`) — see the assertions table above; use `@BeforeEach` if several tests need the same setup.

JaCoCo counters (denominators).

- **Statement/line:** denominator = executable lines/instructions; a line covered if any instruction ran.
- **Branch:** denominator = branch outcomes = **2 per decision**. JaCoCo counts at **bytecode** → each atomic boolean in `&&/|` contributes its own pair. `if (A) → 2; if (A && B) → 4`.

Worked example. `f(y){ if(y>0) return 2*y; if(y<0) return -3*y; return 0; }` with tests `f(5)=10`, `f(-2)=6`, `f(0)=0`:

- `y>0→y>=0`: at `y=0` both return 0 ⇒ **equivalent**. `y<0→y<=0`: at `y=0` both return 0 ⇒ **equivalent**.
- `2*y→2/y`: `f(5) → 2/5=0≠10 ⇒ killed`. `-3*y→-3/y`: `f(-2) → -3/-2=1≠6 ⇒ killed`.
- `N=4`, `E=2`, `D=2` → **score $100 \times 2 / (4 - 2) = 100\%$** . State the survivors are equivalent; don't "fix" them.

Exam patterns & gotchas.

- **CONDITIONALS_BOUNDARY** at an **untested boundary value** → **often equivalent**. Always test the equality case to decide.
- `a*b→b*a`, `x+0→x`, mutating unreachable code ⇒ equivalent.
- **100% branch coverage** ⇒ **all mutants killed**. The *oracle* is the assertion that decides pass/fail; a **weak oracle** runs the mutated line and the mutant even computes a *different* value, but the assertion is too loose to notice — so the mutant survives despite full coverage. Typical sub-question: *"write a test that covers the mutated statement yet still passes on the mutant."* Example — `int f(int x){ return x*2; }` with mutant `*→+` (so `f(3)` is 6 in the original, 5 in the mutant). The test `assertTrue(f(3) > 0)` executes the line but only checks the sign — `6>0` and `5>0` both hold ⇒ mutant survives. Fix: `assertEquals(6, f(3))`, which sees `5 ≠ 6` ⇒ killed.
- Always put the **delta on double** `assertEquals`. Score denominator is **N – E**, never N.